

BFS Traversal

Step 1: Start at E

Legend



Complete



Processing



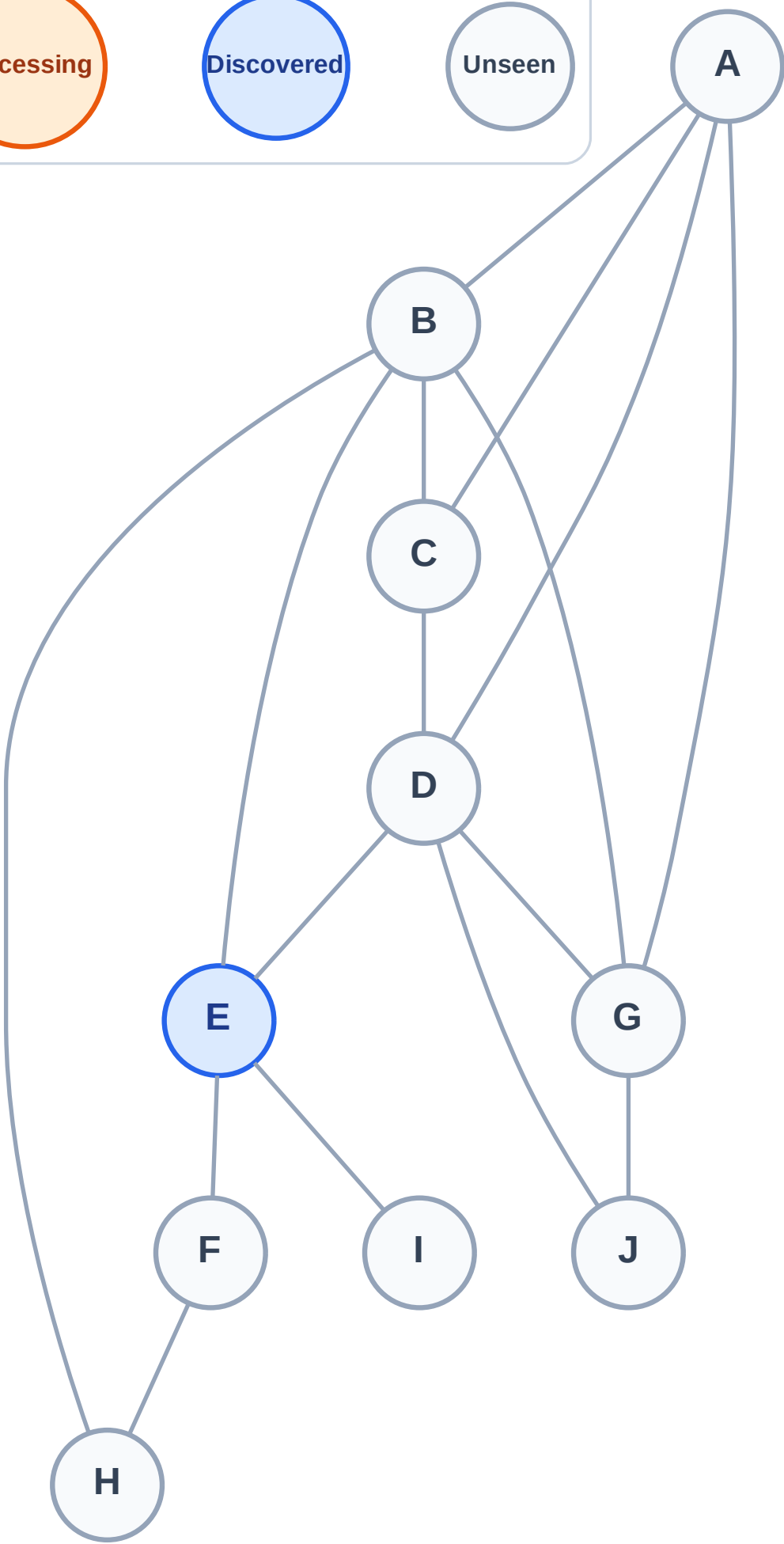
Discovered



Unseen

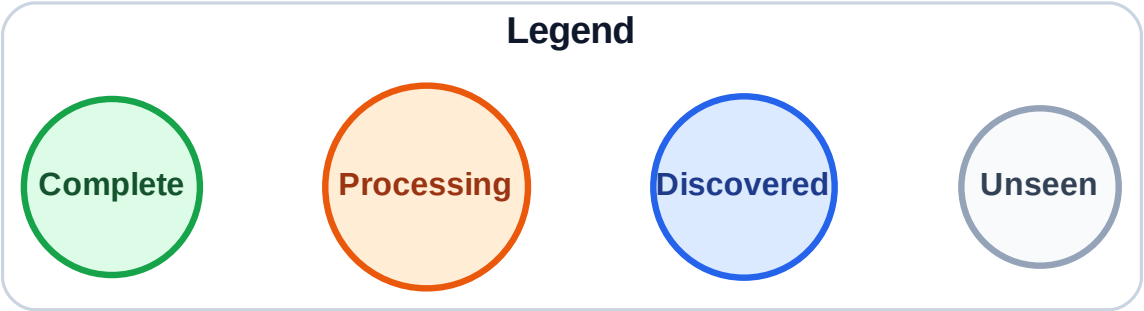
Queue: E

Visited: (none)

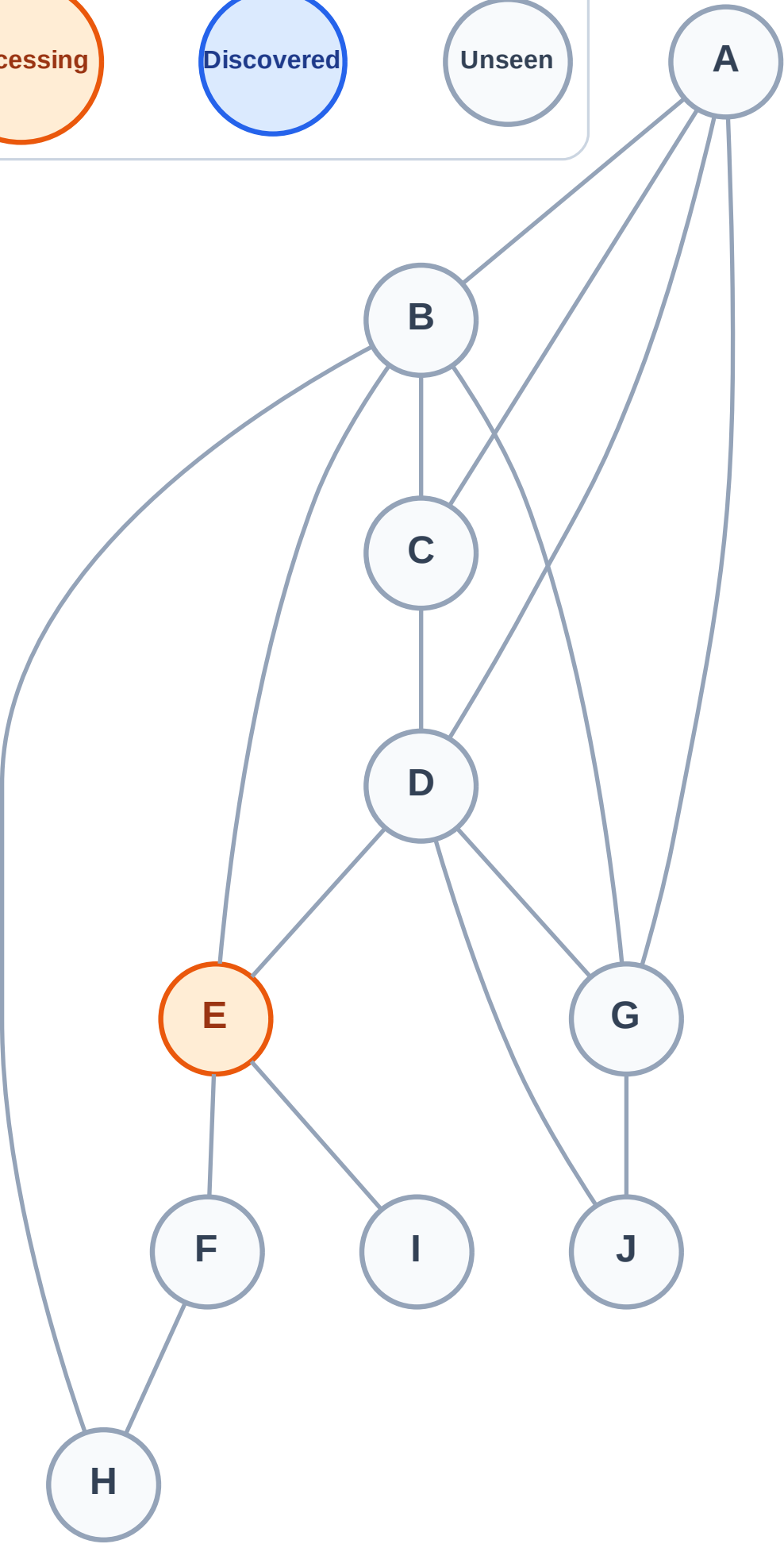


BFS Traversal

Step 2: Process node E



Queue: (empty)
Visited: (none)



BFS Traversal

Step 3: Discover B, D, F, I from E

Legend

Complete

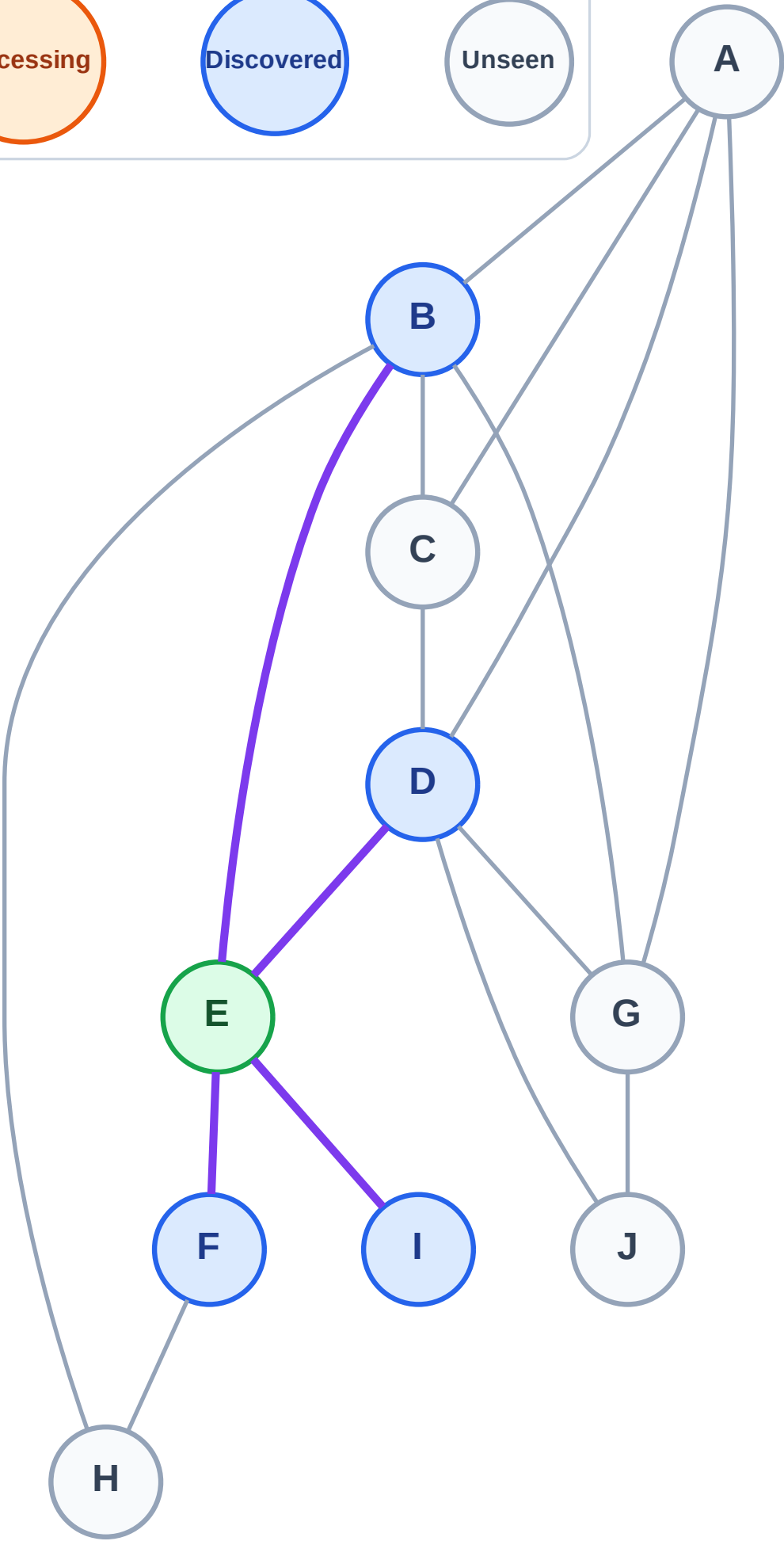
Processing

Discovered

Unseen

Queue: B → D → F → I

Visited: E



BFS Traversal

Step 4: Process node B

Legend

Complete

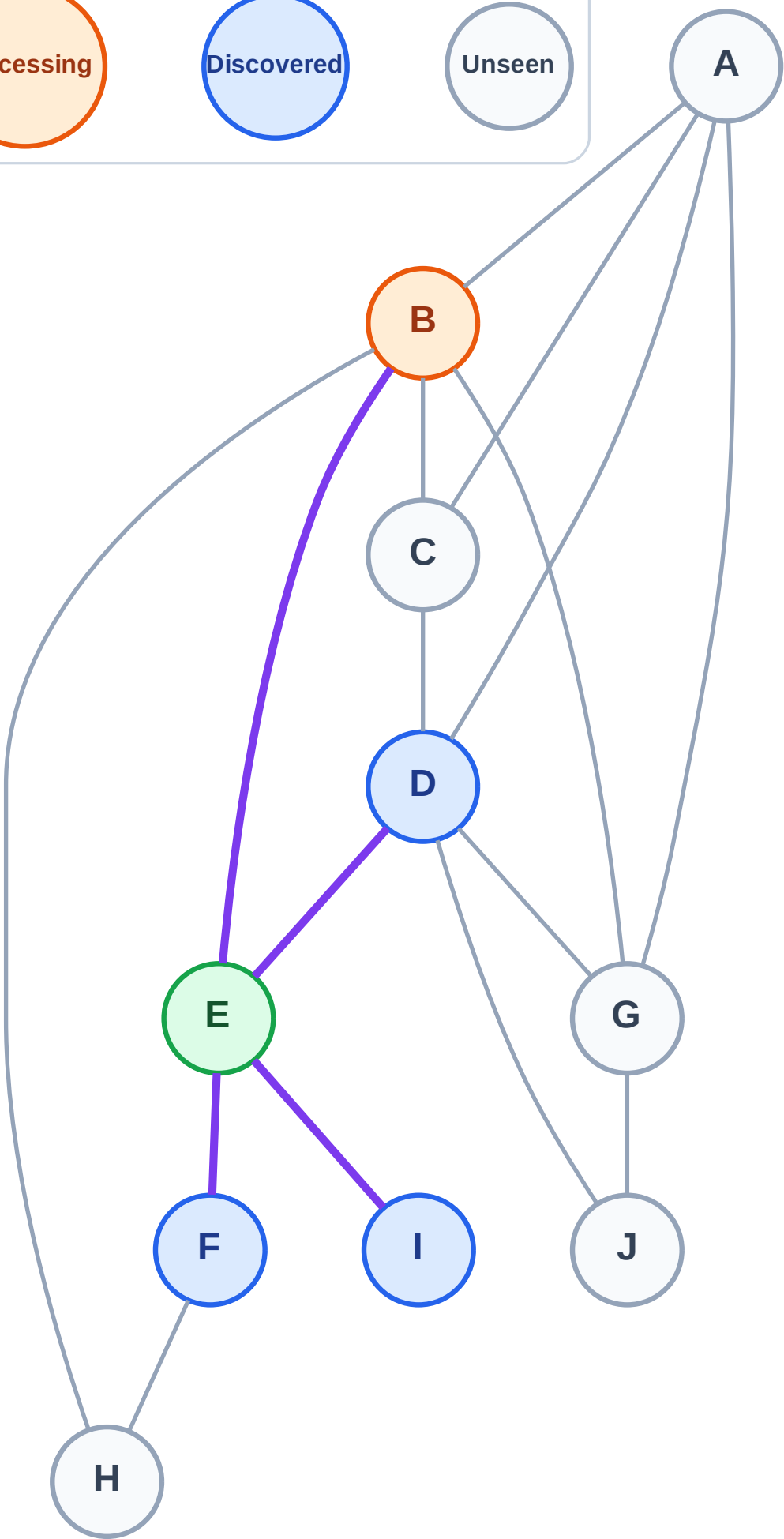
Processing

Discovered

Unseen

Queue: D → F → I

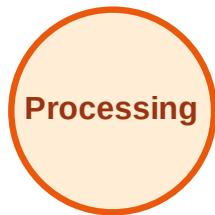
Visited: E



BFS Traversal

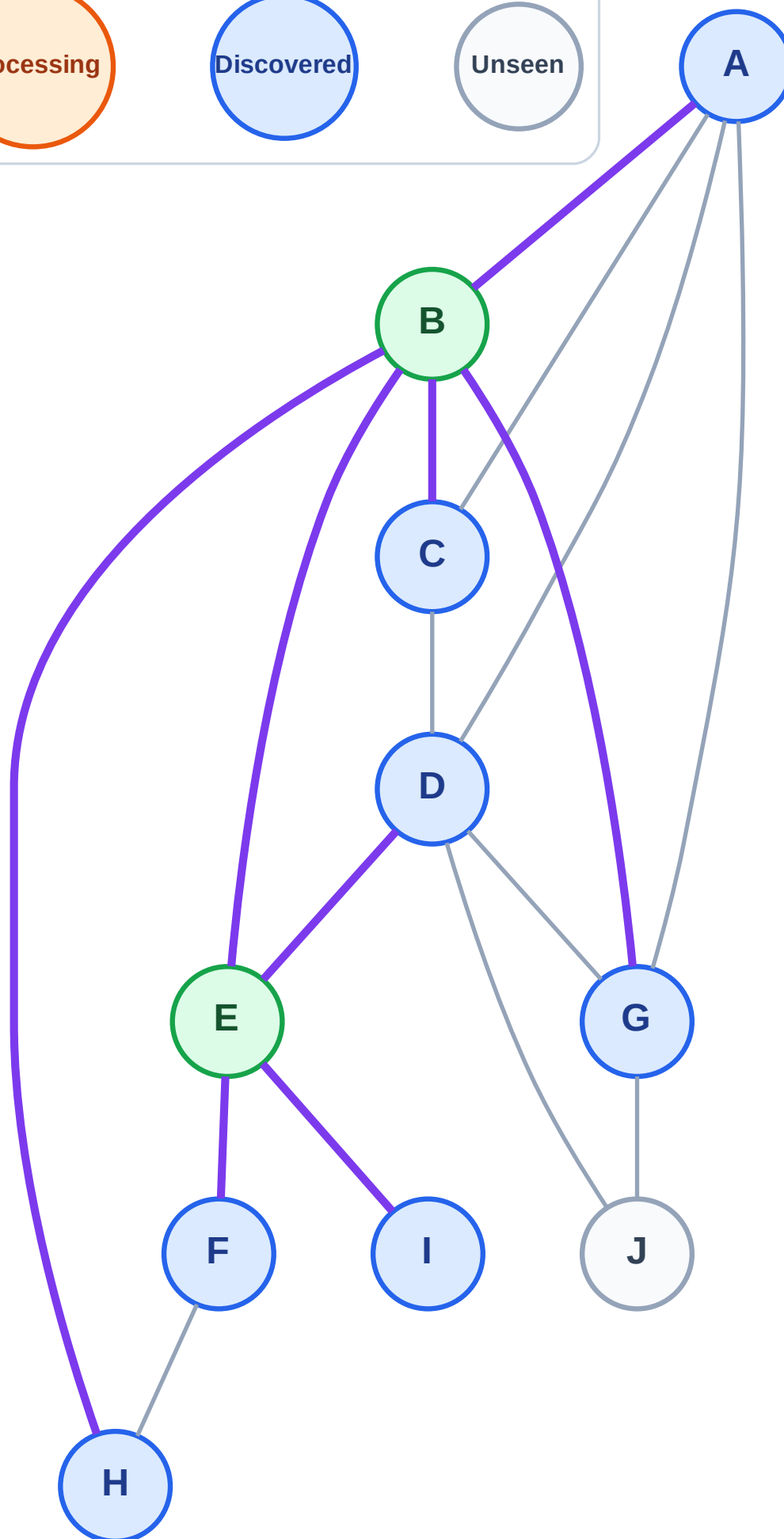
Step 5: Discover A, C, G, H from B

Legend



Queue: D → F → I → A → C → G → H

Visited: B, E



BFS Traversal

Step 6: Process node D

Legend

Complete

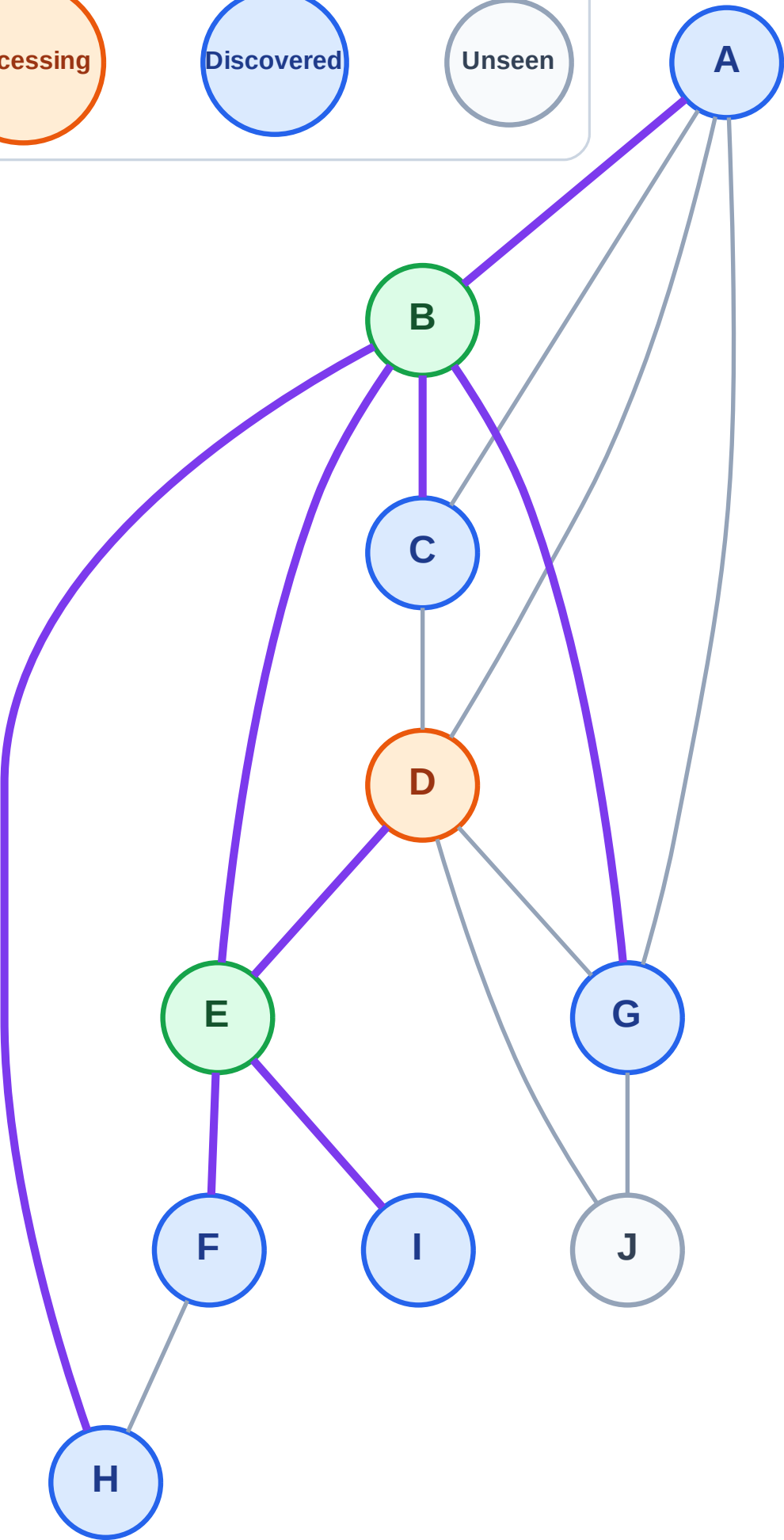
Processing

Discovered

Unseen

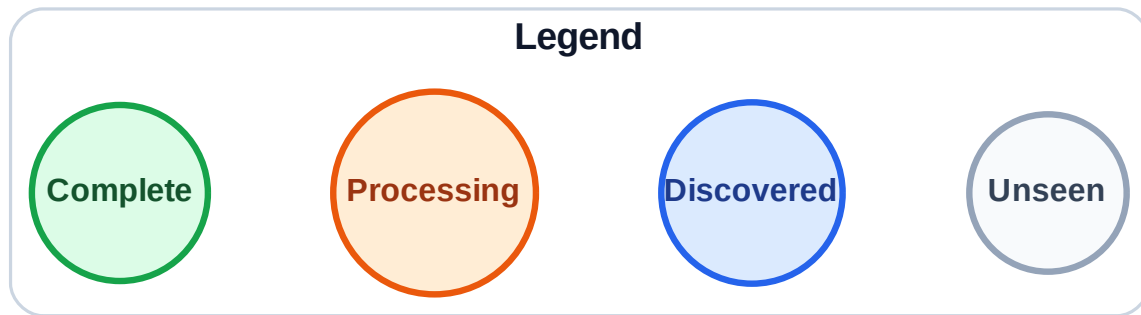
Queue: F → I → A → C → G → H

Visited: B, E



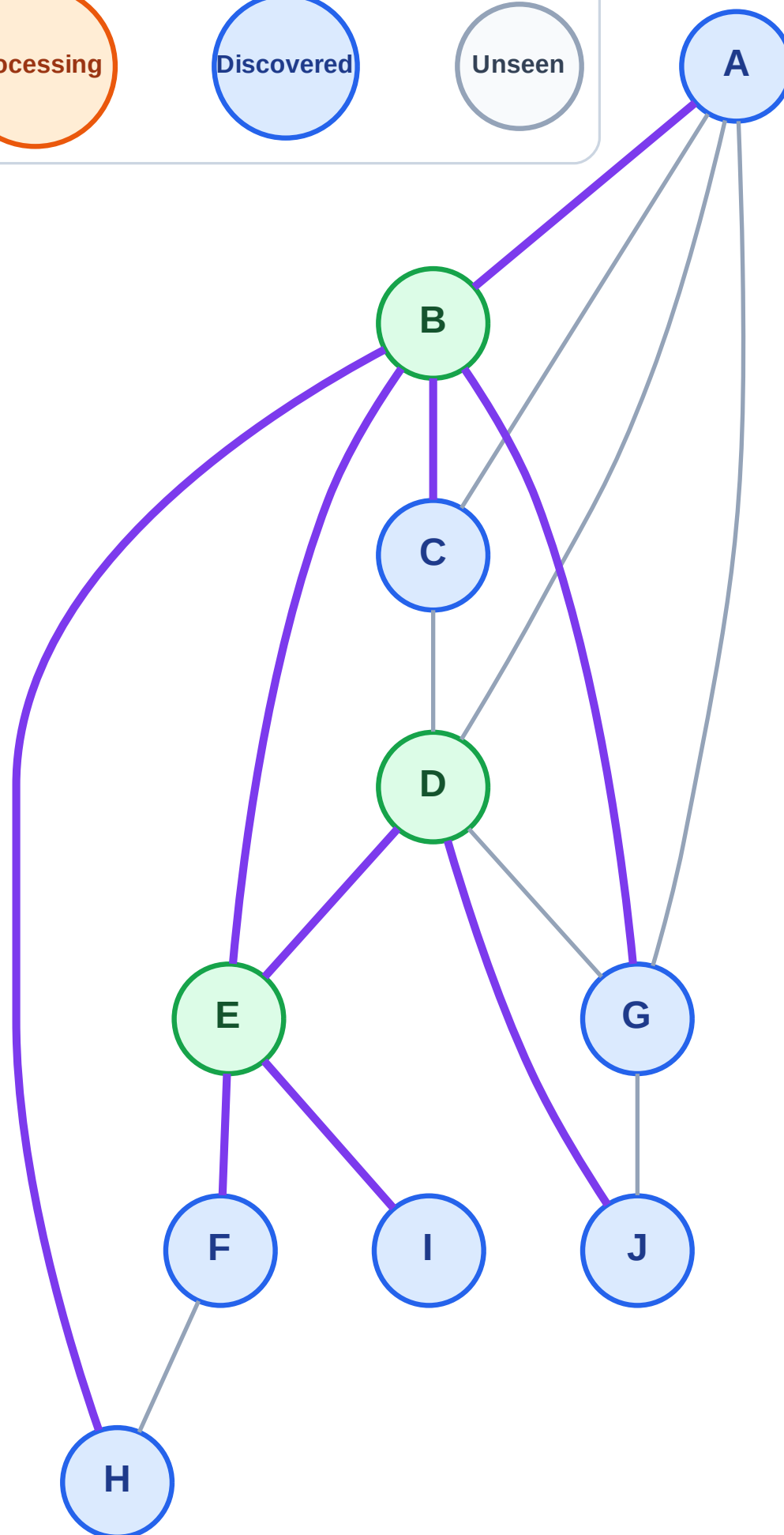
BFS Traversal

Step 7: Discover J from D



Queue: F → I → A → C → G → H → J

Visited: B, D, E

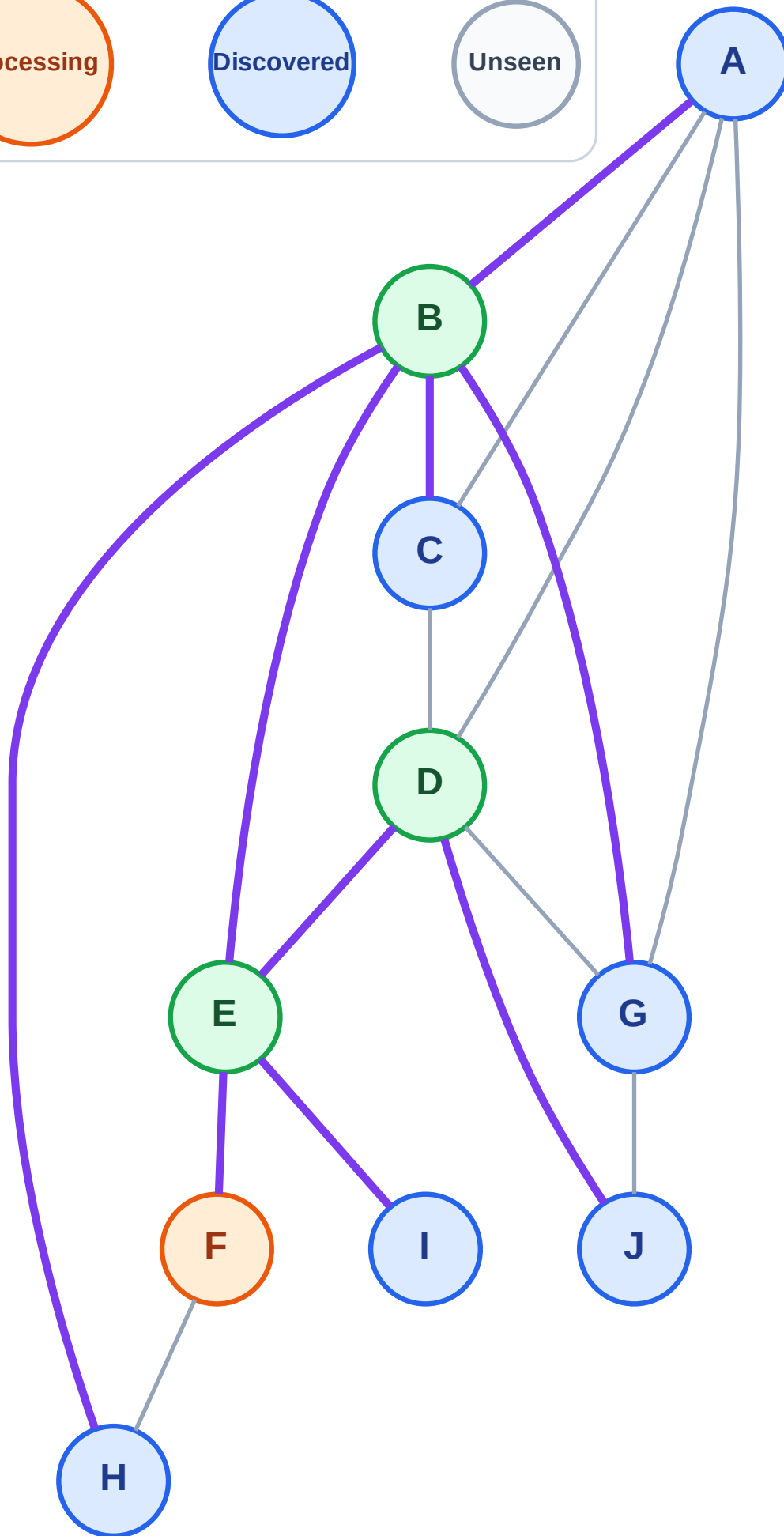
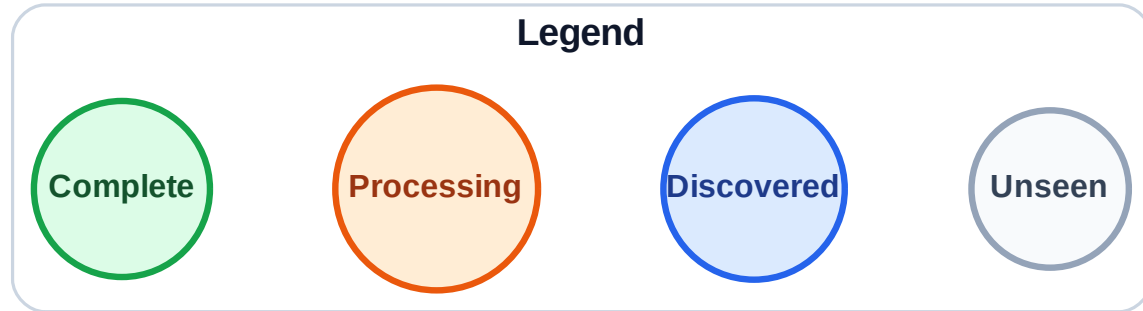


BFS Traversal

Step 8: Process node F

Queue: I → A → C → G → H → J

Visited: B, D, E



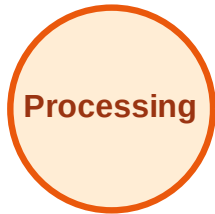
BFS Traversal

Step 9: No new neighbors from F

Legend



Complete



Processing



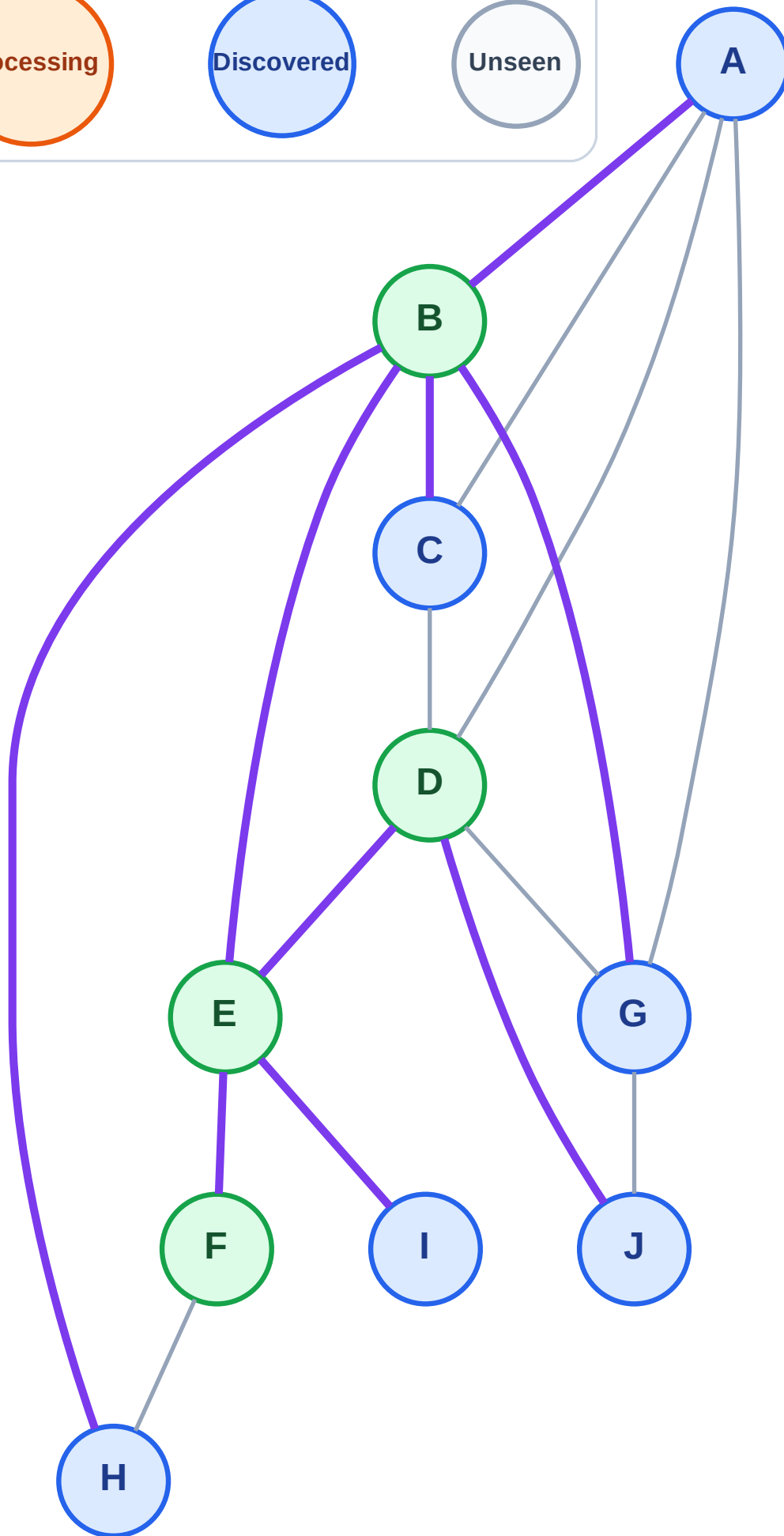
Discovered



Unseen

Queue: I → A → C → G → H → J

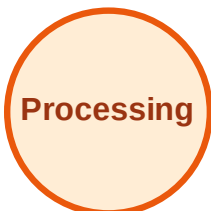
Visited: B, D, E, F



BFS Traversal

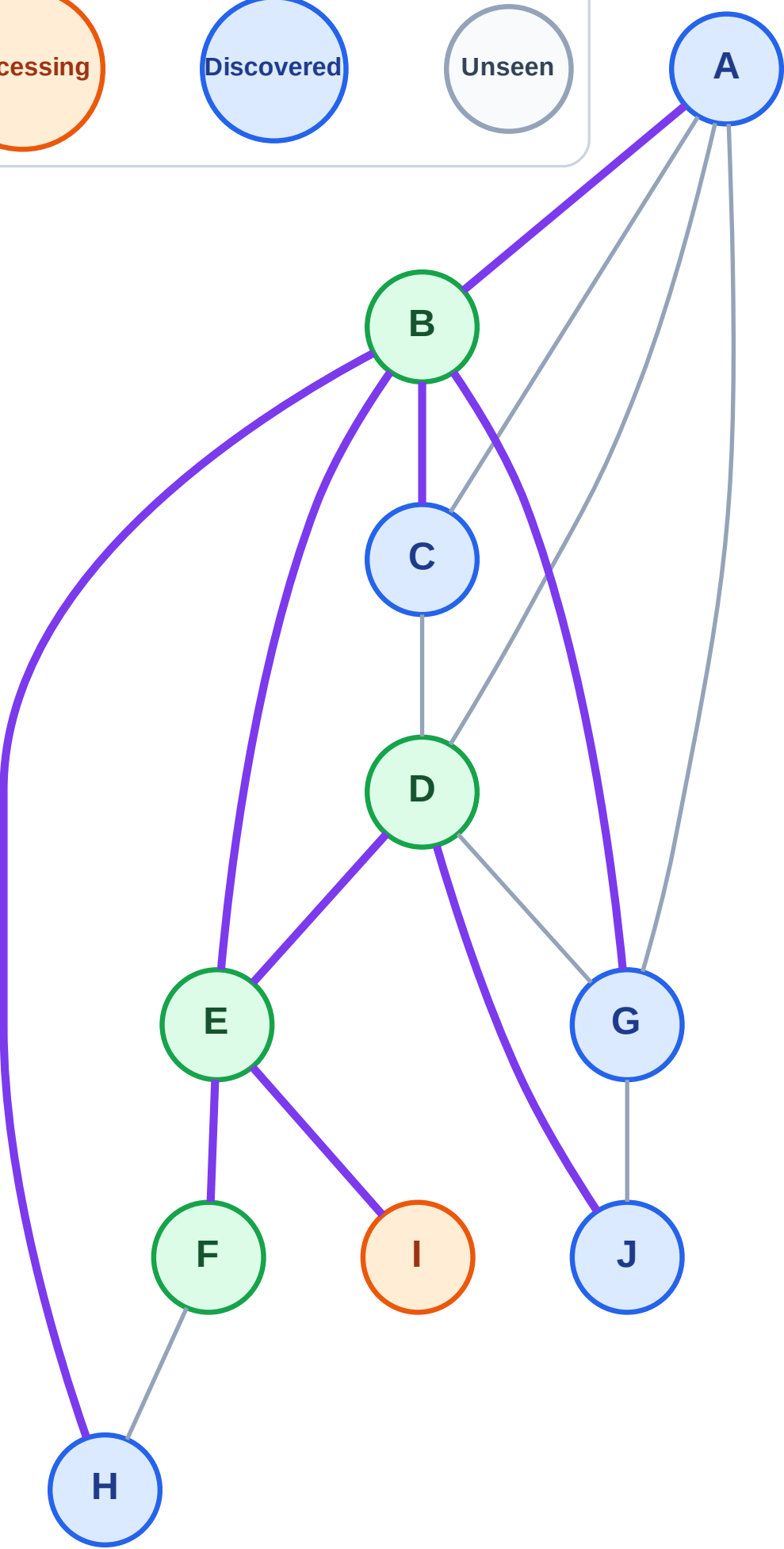
Step 10: Process node I

Legend



Queue: A → C → G → H → J

Visited: B, D, E, F



BFS Traversal

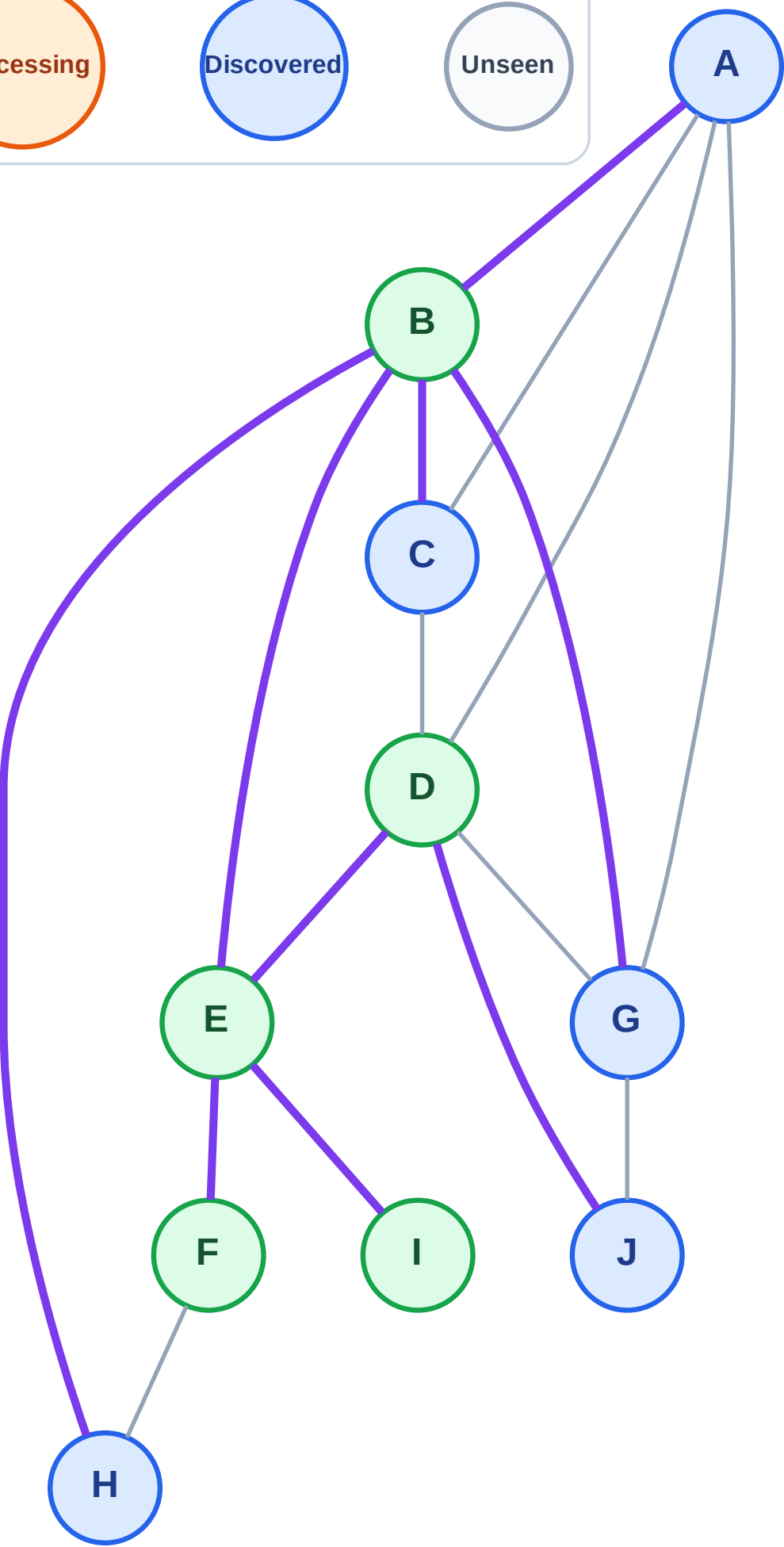
Step 11: No new neighbors from I

Legend



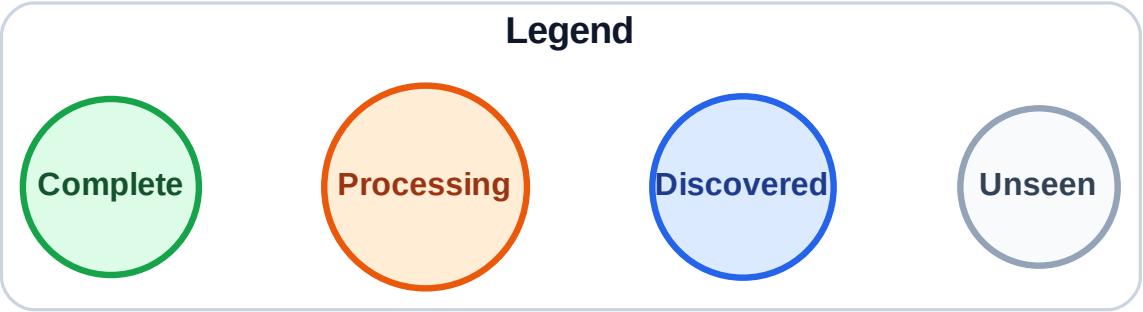
Queue: A → C → G → H → J

Visited: B, D, E, F, I



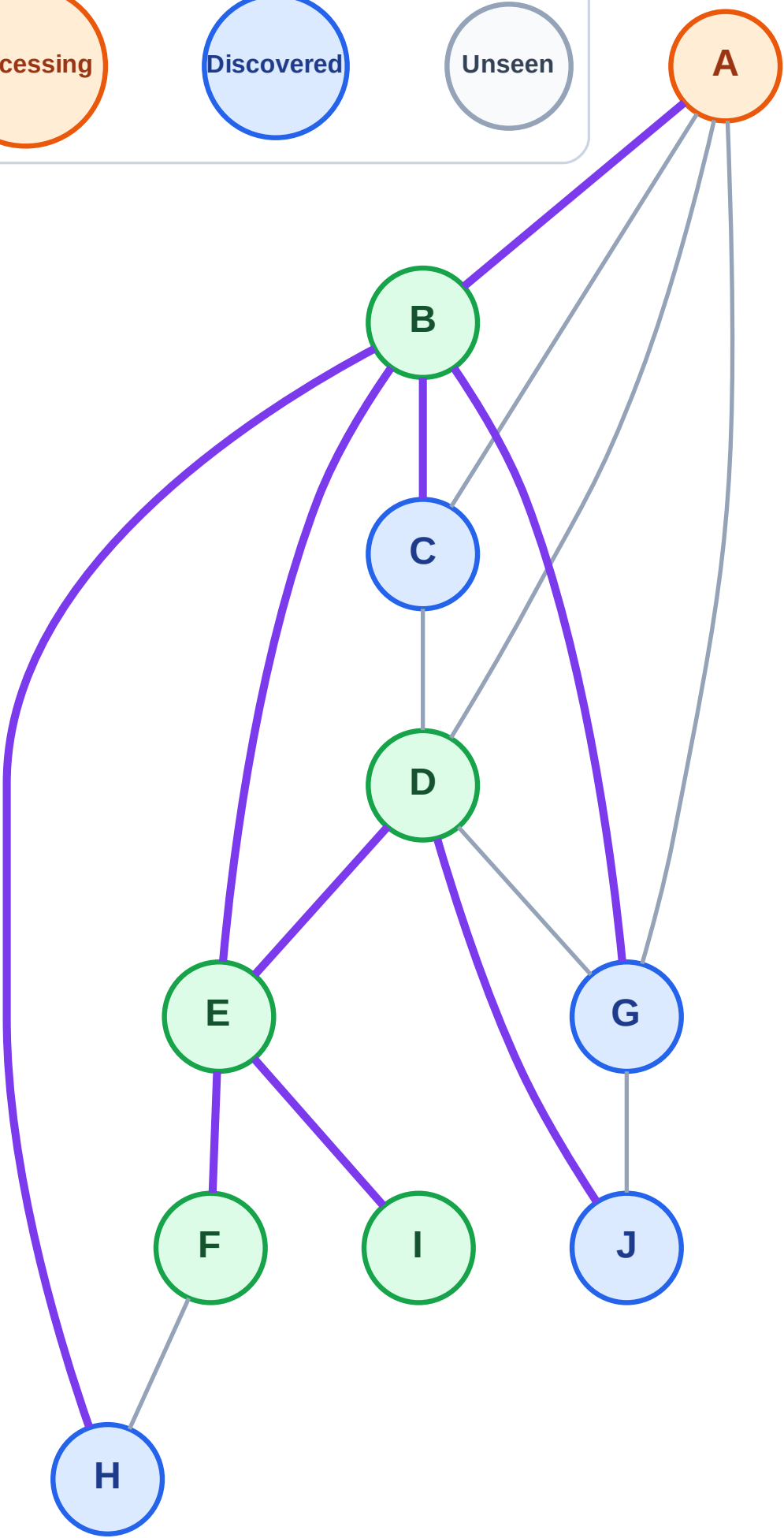
BFS Traversal

Step 12: Process node A



Queue: C → G → H → J

Visited: B, D, E, F, I



BFS Traversal

Step 13: No new neighbors from A

Legend

Complete

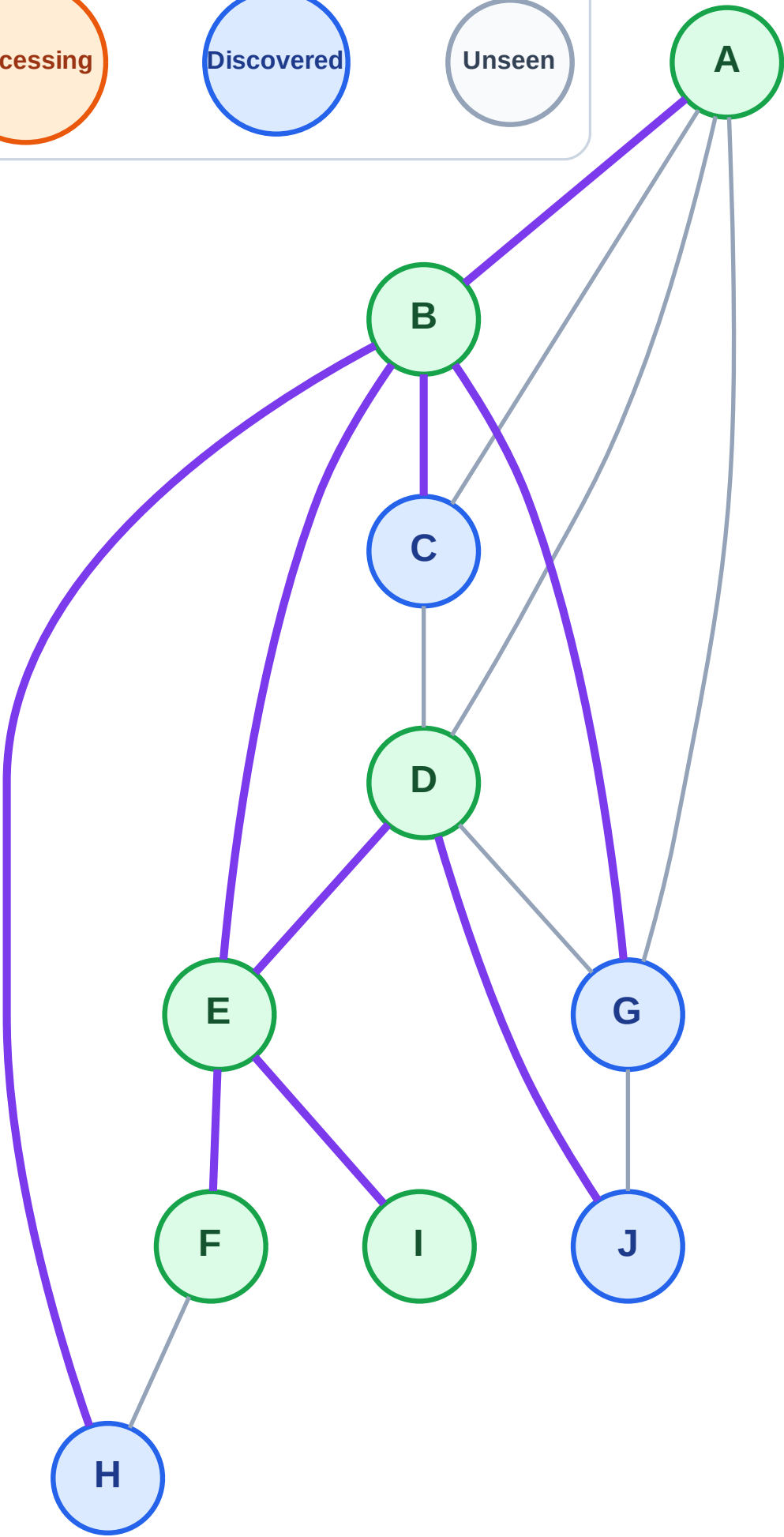
Processing

Discovered

Unseen

Queue: C → G → H → J

Visited: A, B, D, E, F, I



BFS Traversal

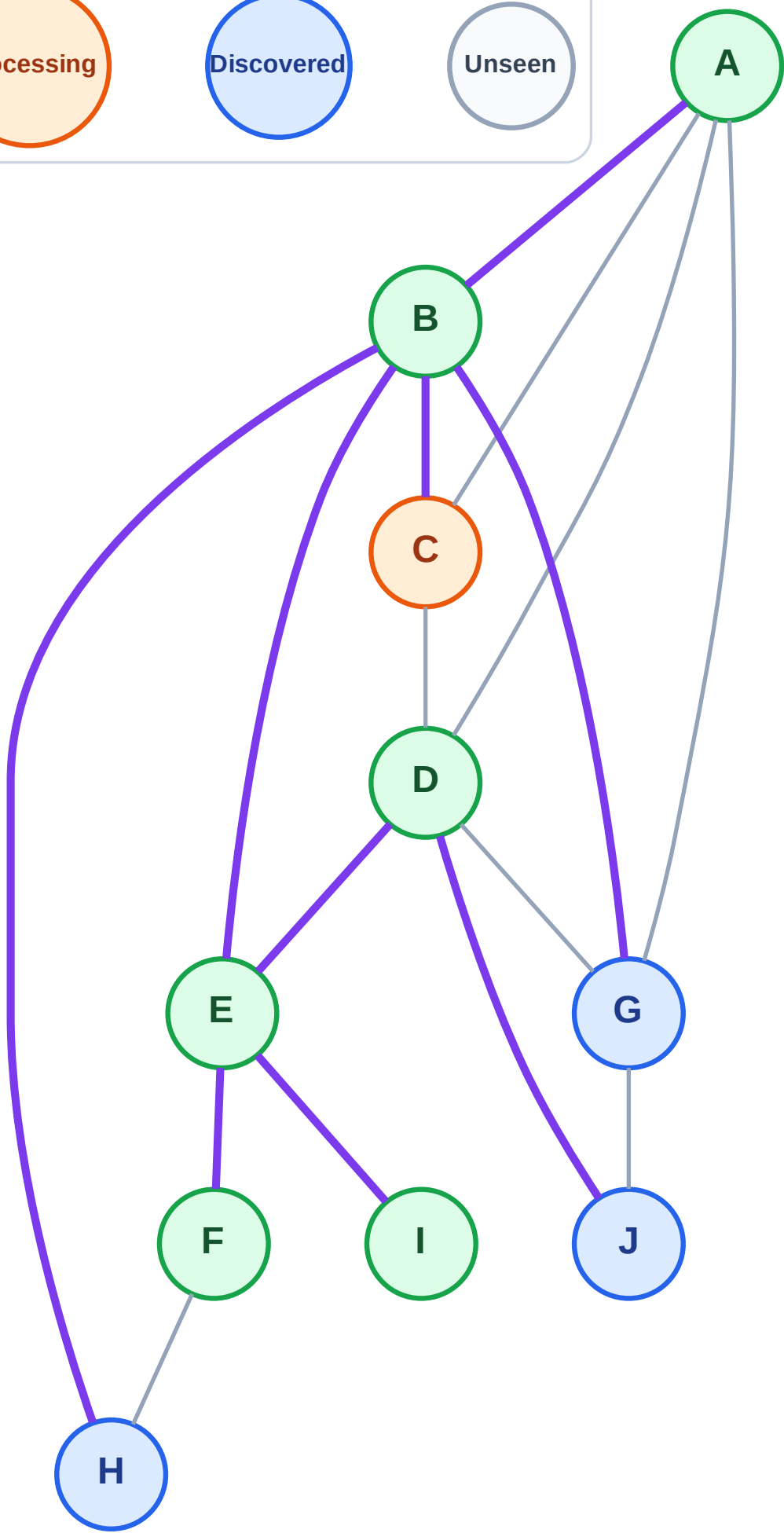
Step 14: Process node C

Legend



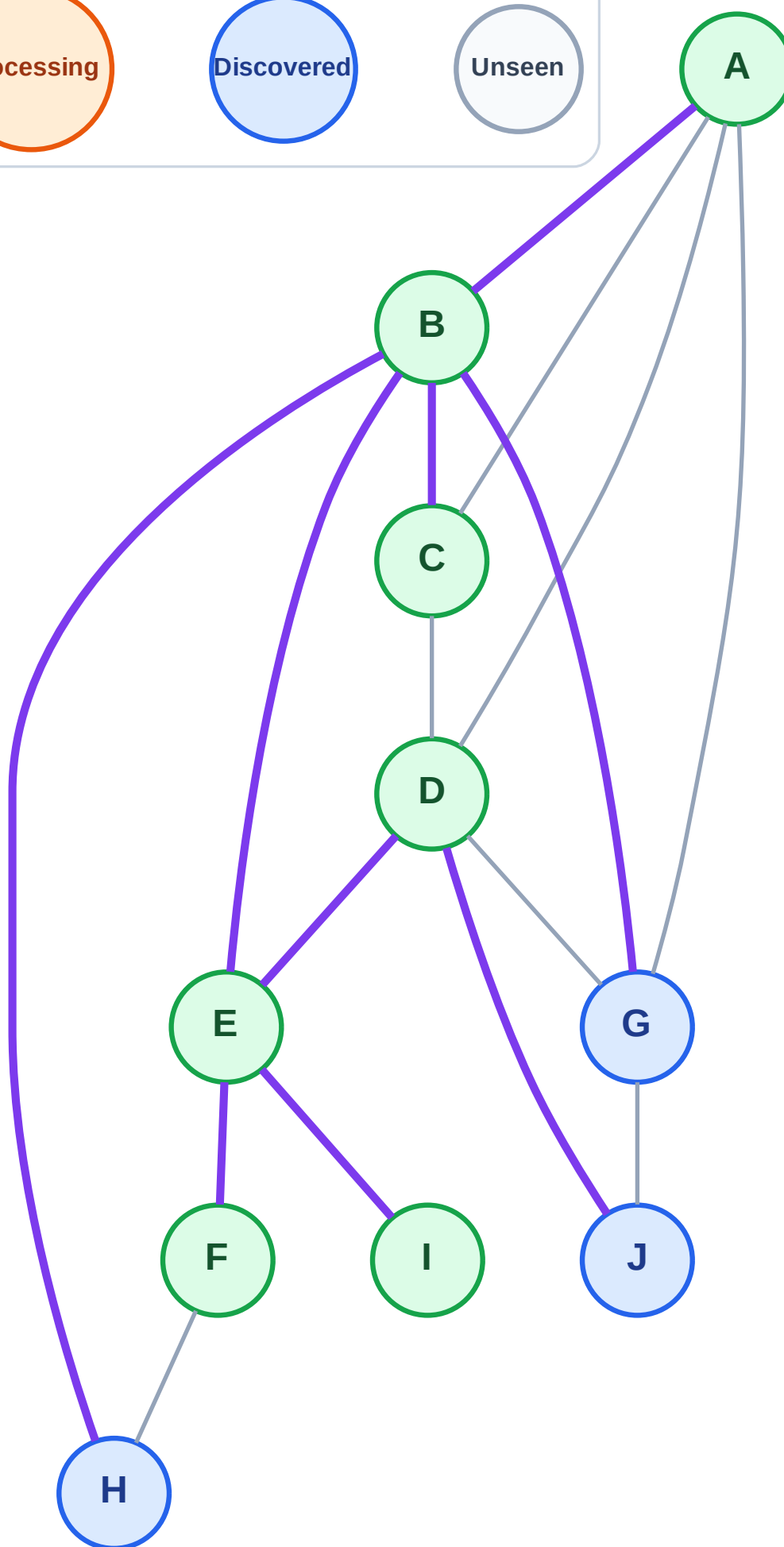
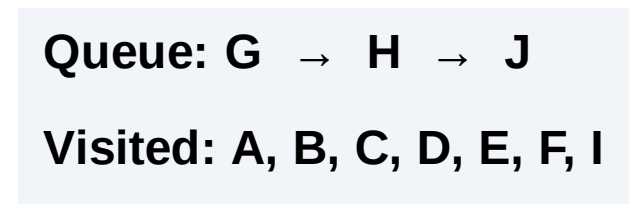
Queue: G → H → J

Visited: A, B, D, E, F, I



Step 15: No new neighbors from C

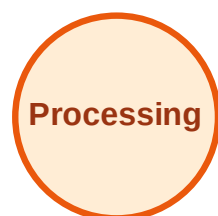
Step 15: No new neighbors from C



BFS Traversal

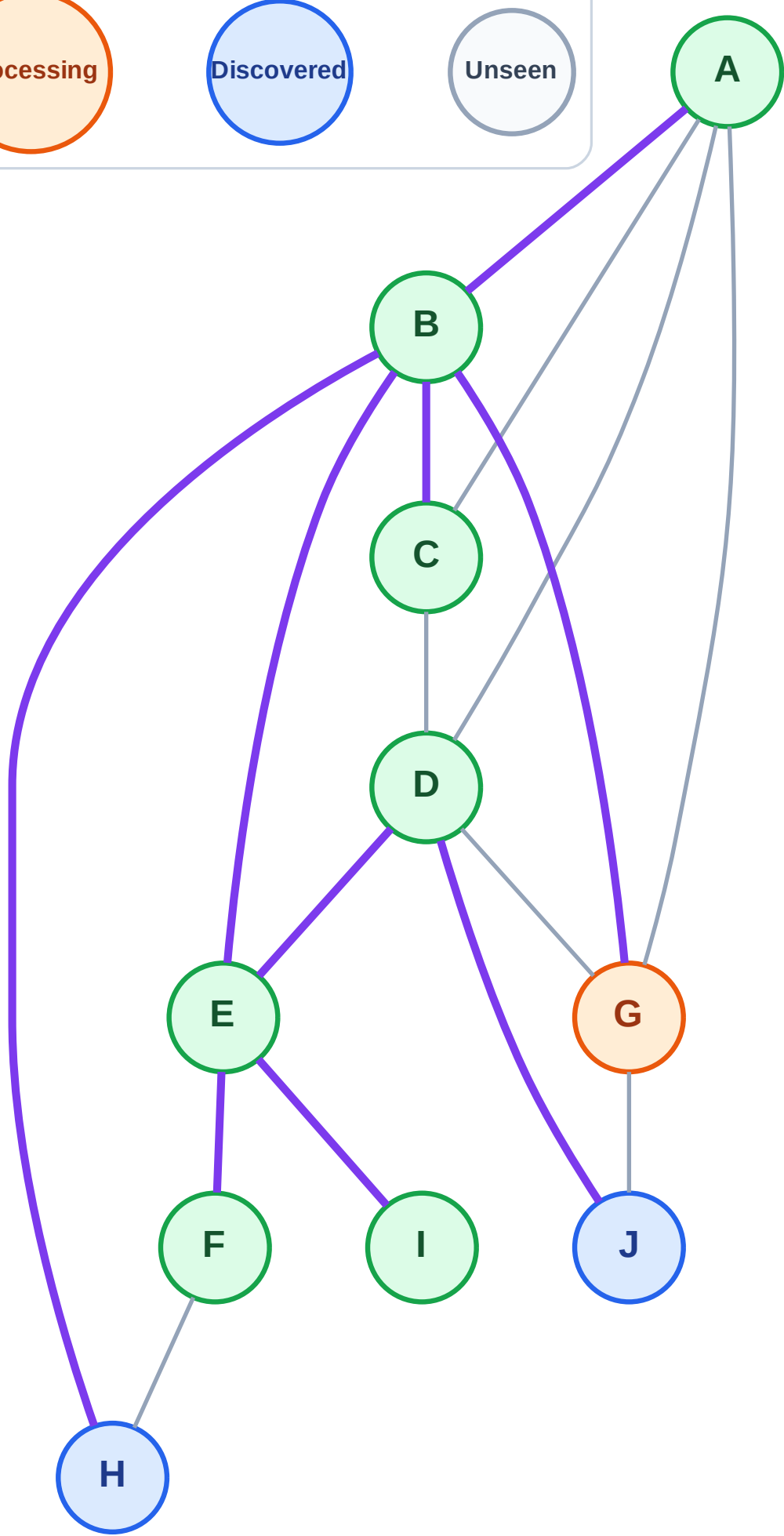
Step 16: Process node G

Legend



Queue: H → J

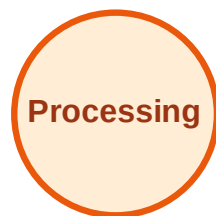
Visited: A, B, C, D, E, F, I



BFS Traversal

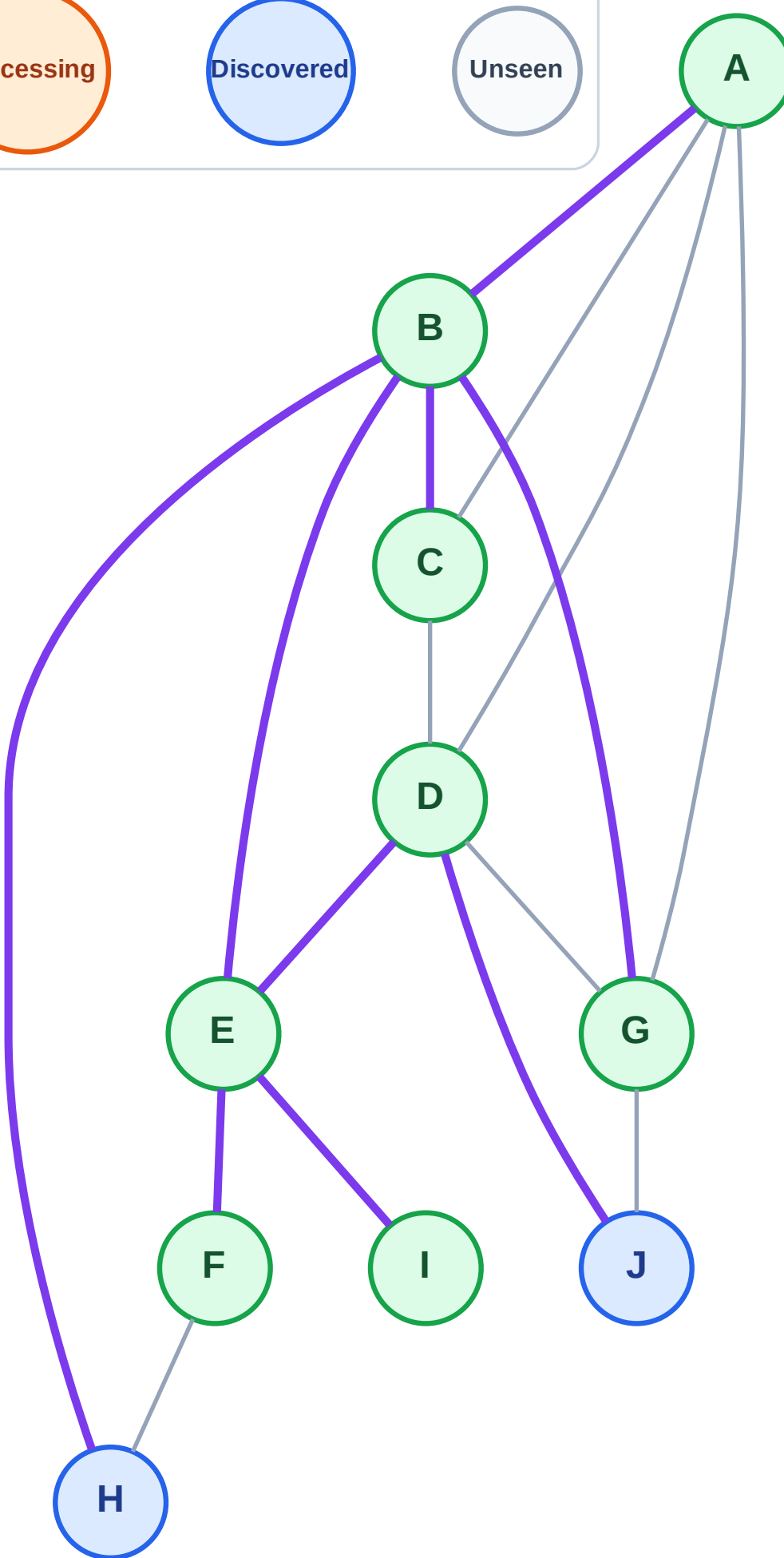
Step 17: No new neighbors from G

Legend



Queue: H → J

Visited: A, B, C, D, E, F, G, I



BFS Traversal

Step 18: Process node H

Legend

Complete

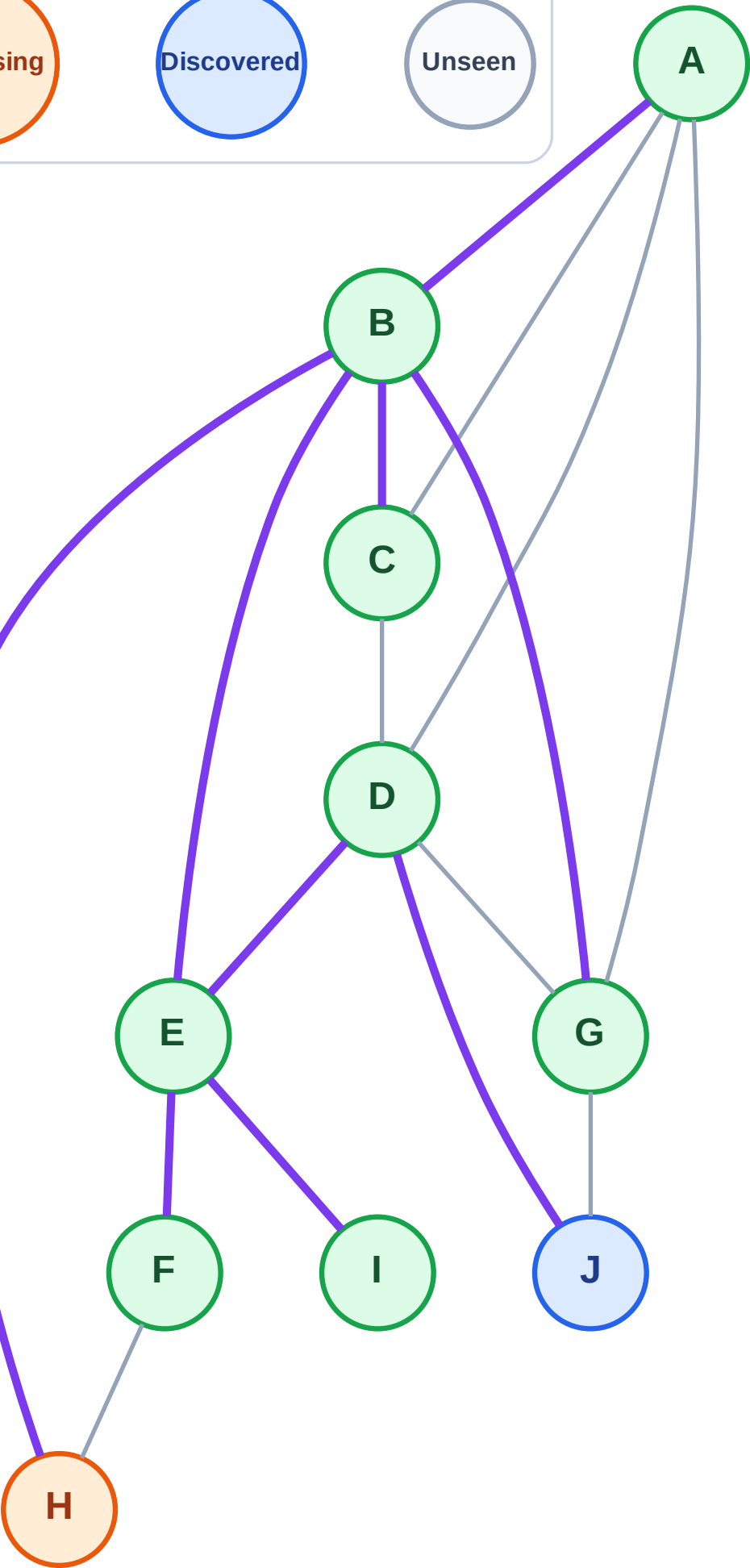
Processing

Discovered

Unseen

Queue: J

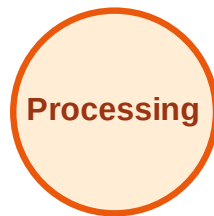
Visited: A, B, C, D, E, F, G, I



BFS Traversal

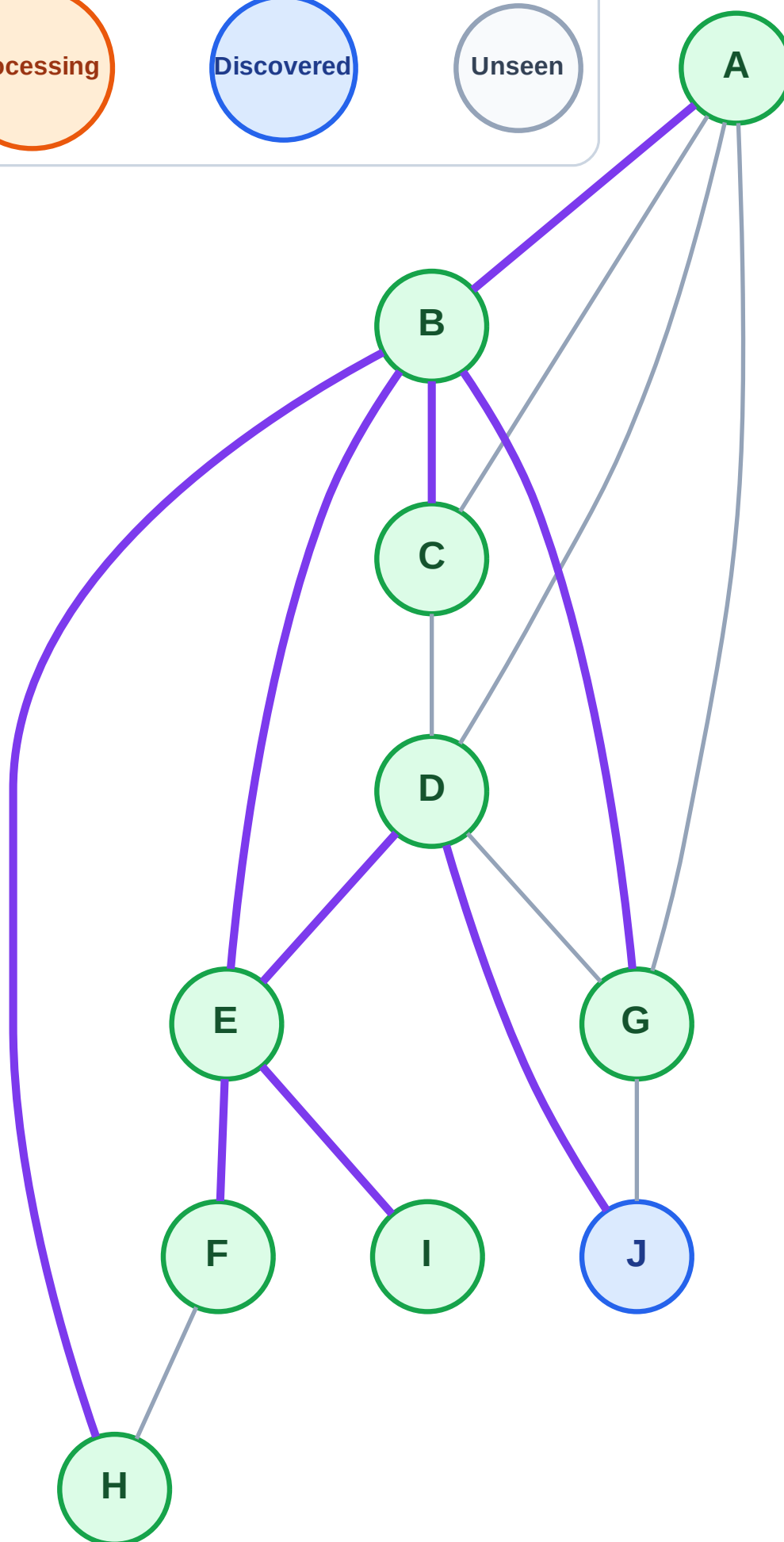
Step 19: No new neighbors from H

Legend



Queue: J

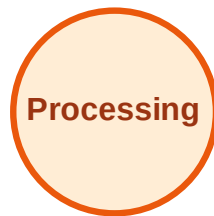
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

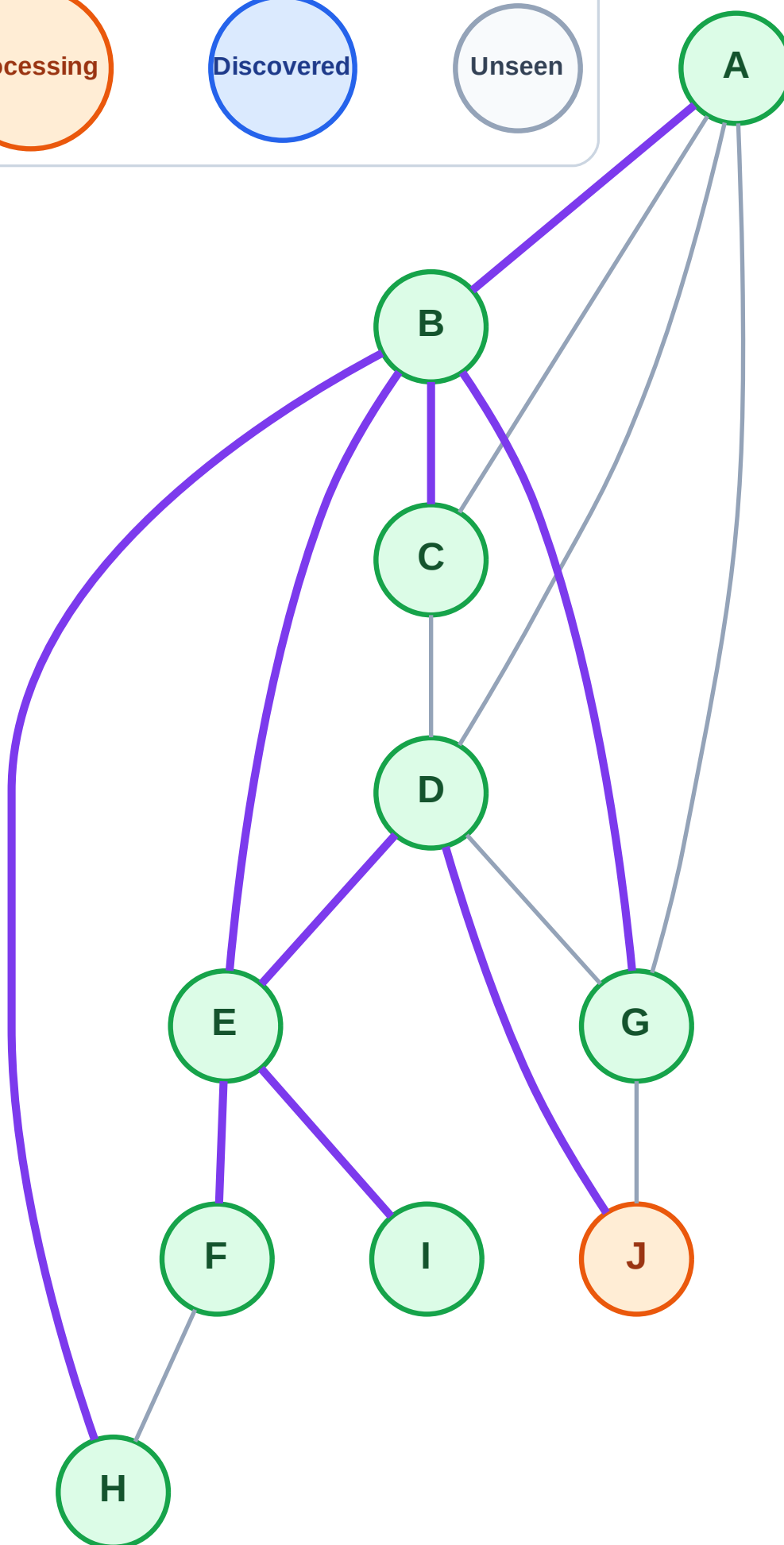
Step 20: Process node J

Legend



Queue: (empty)

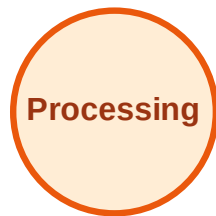
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

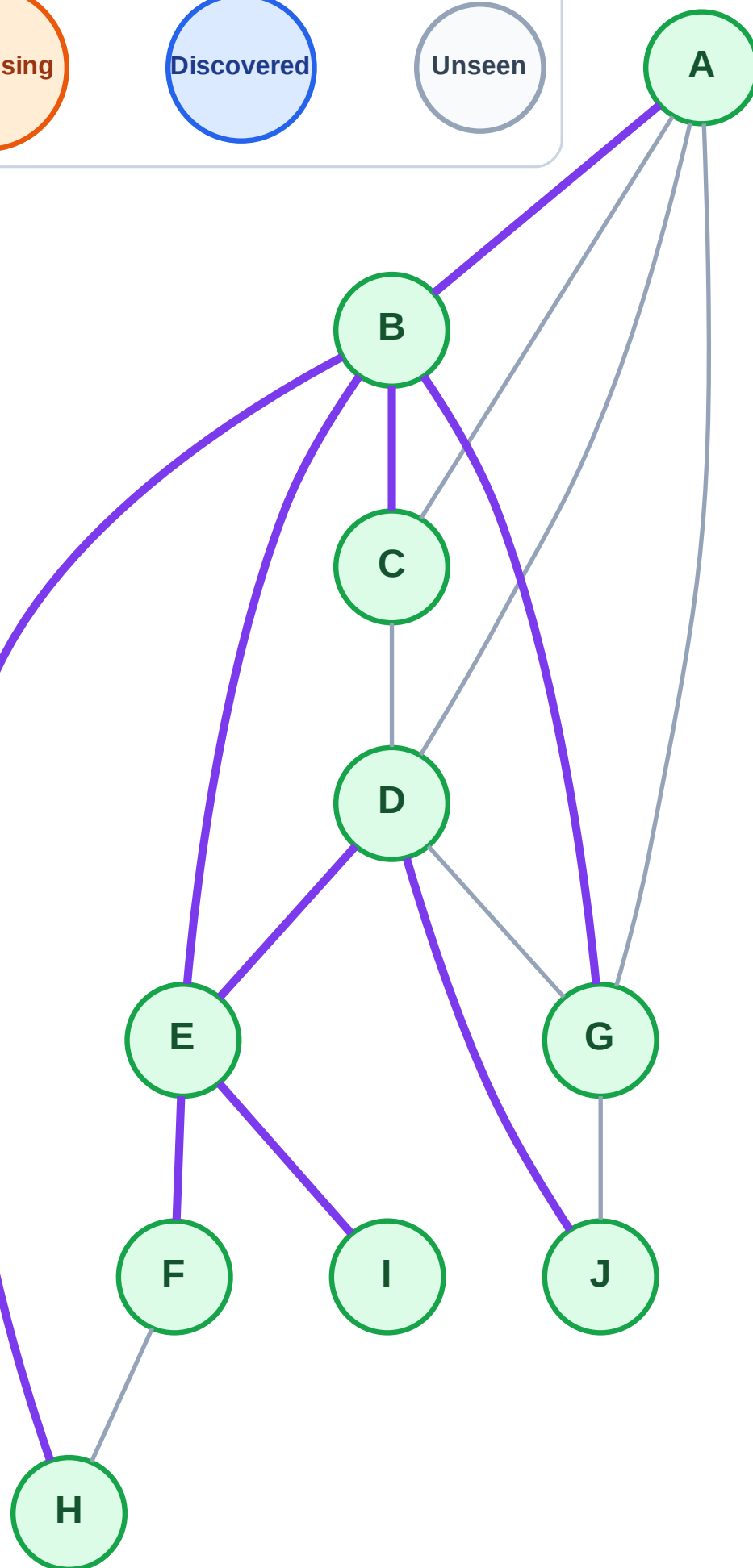
Step 21: No new neighbors from J

Legend



Queue: (empty)

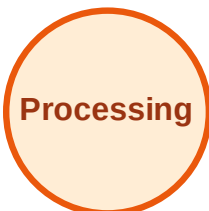
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

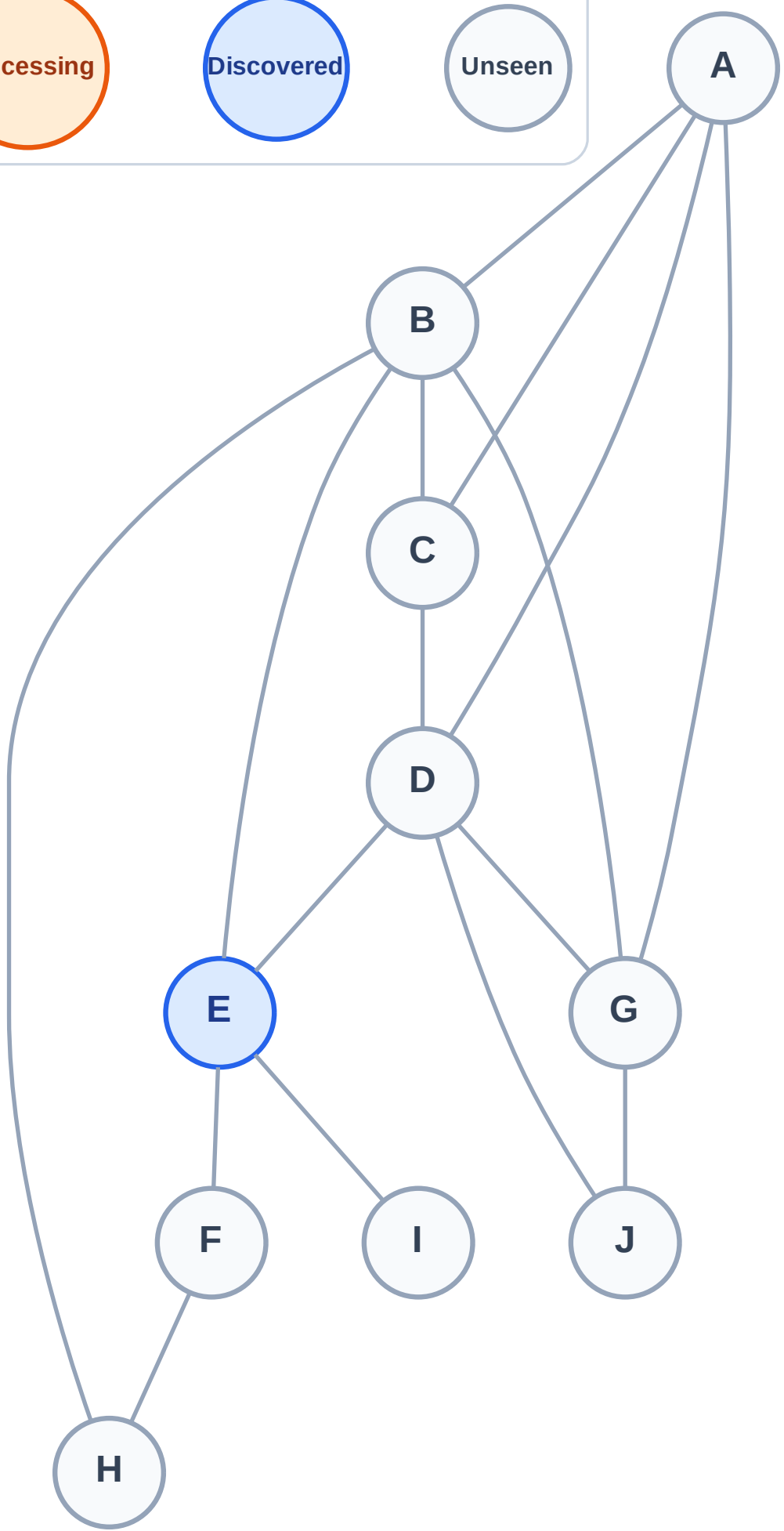
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

Complete

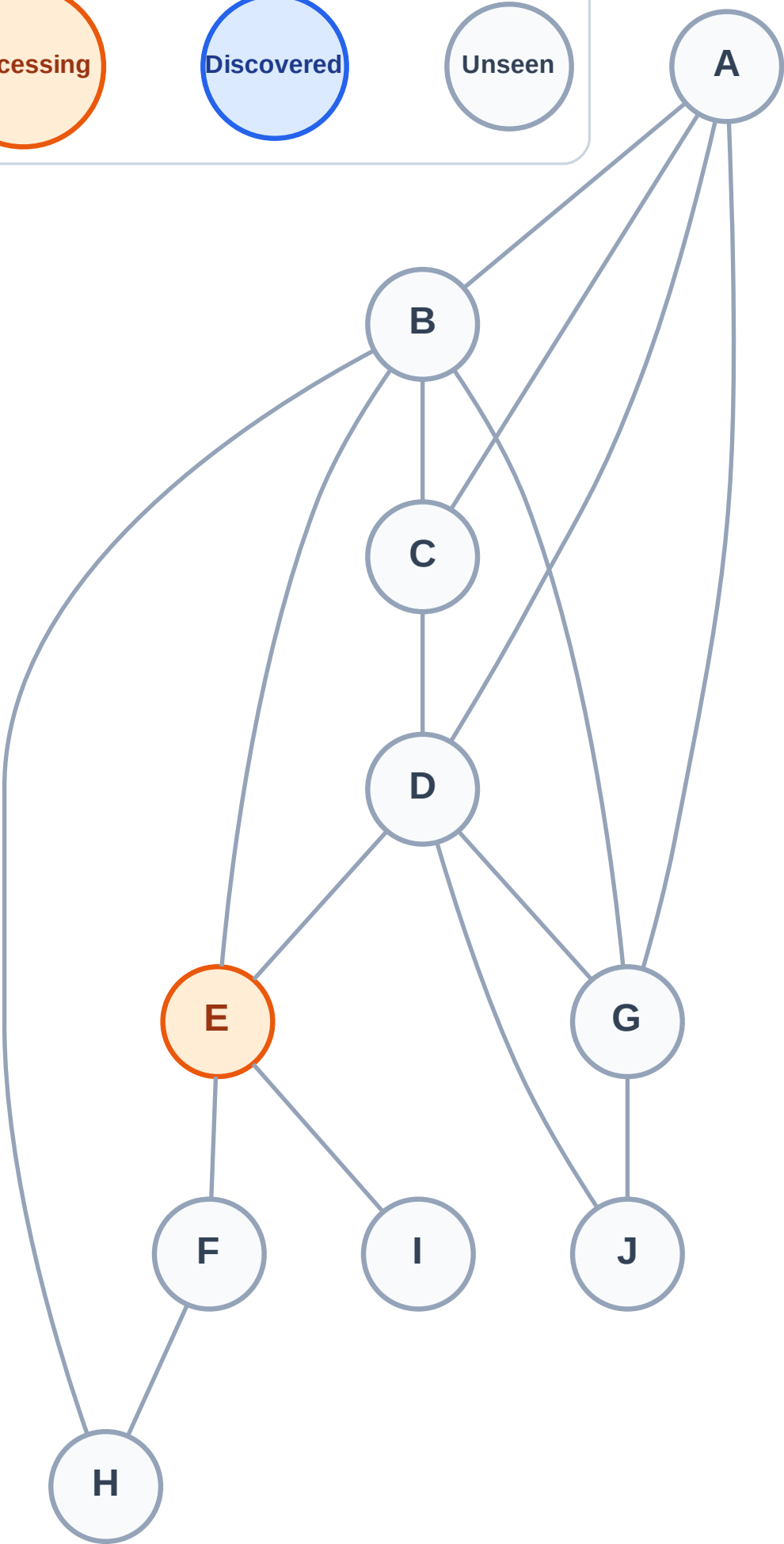
Processing

Discovered

Unseen

Stack (top at right): (empty)

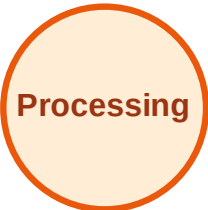
Visited: (none)



DFS Traversal

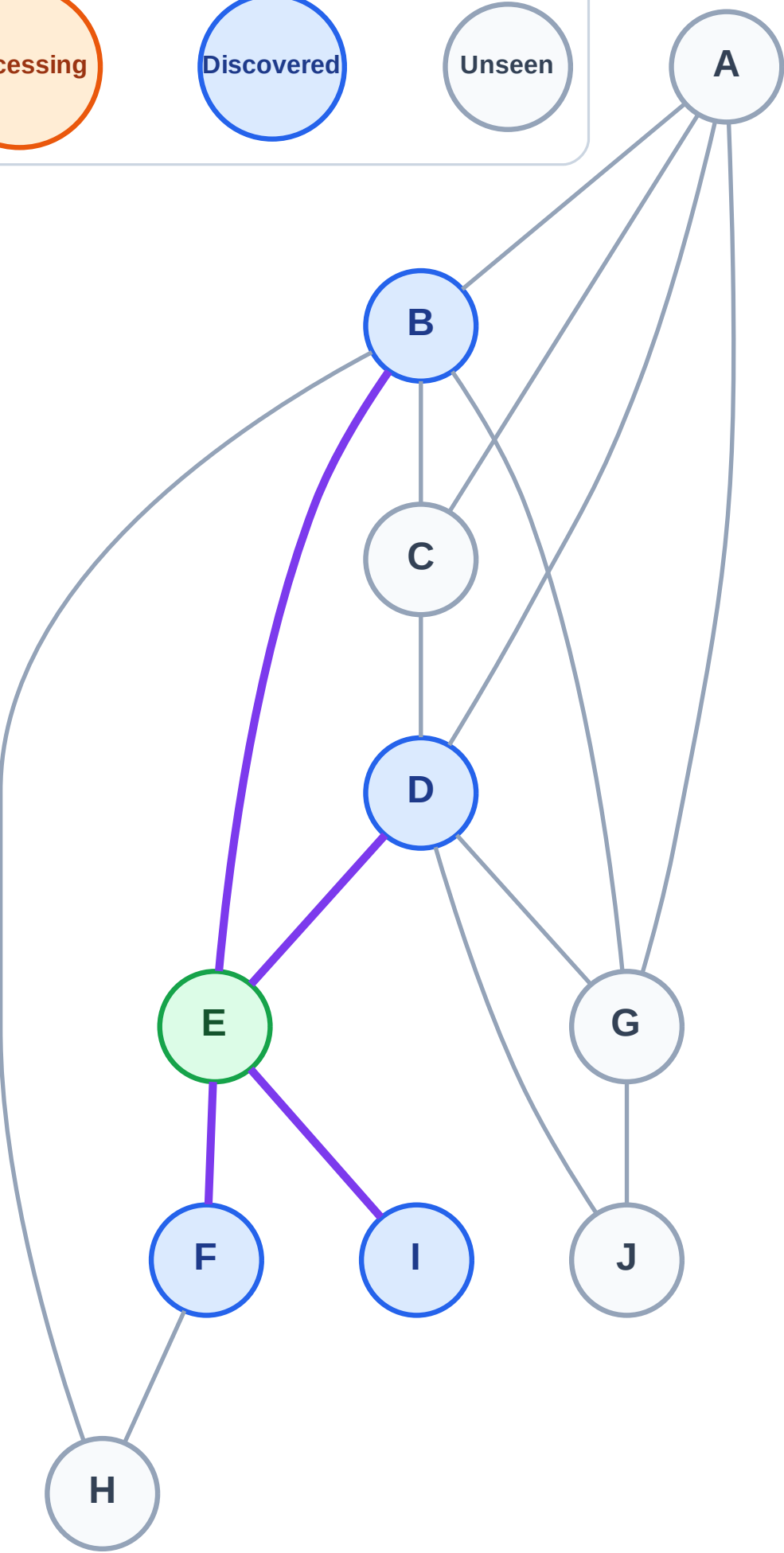
Step 3: Discover I, F, D, B from E

Legend



Stack (top at right): I | F | D | B

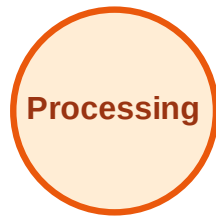
Visited: E



DFS Traversal

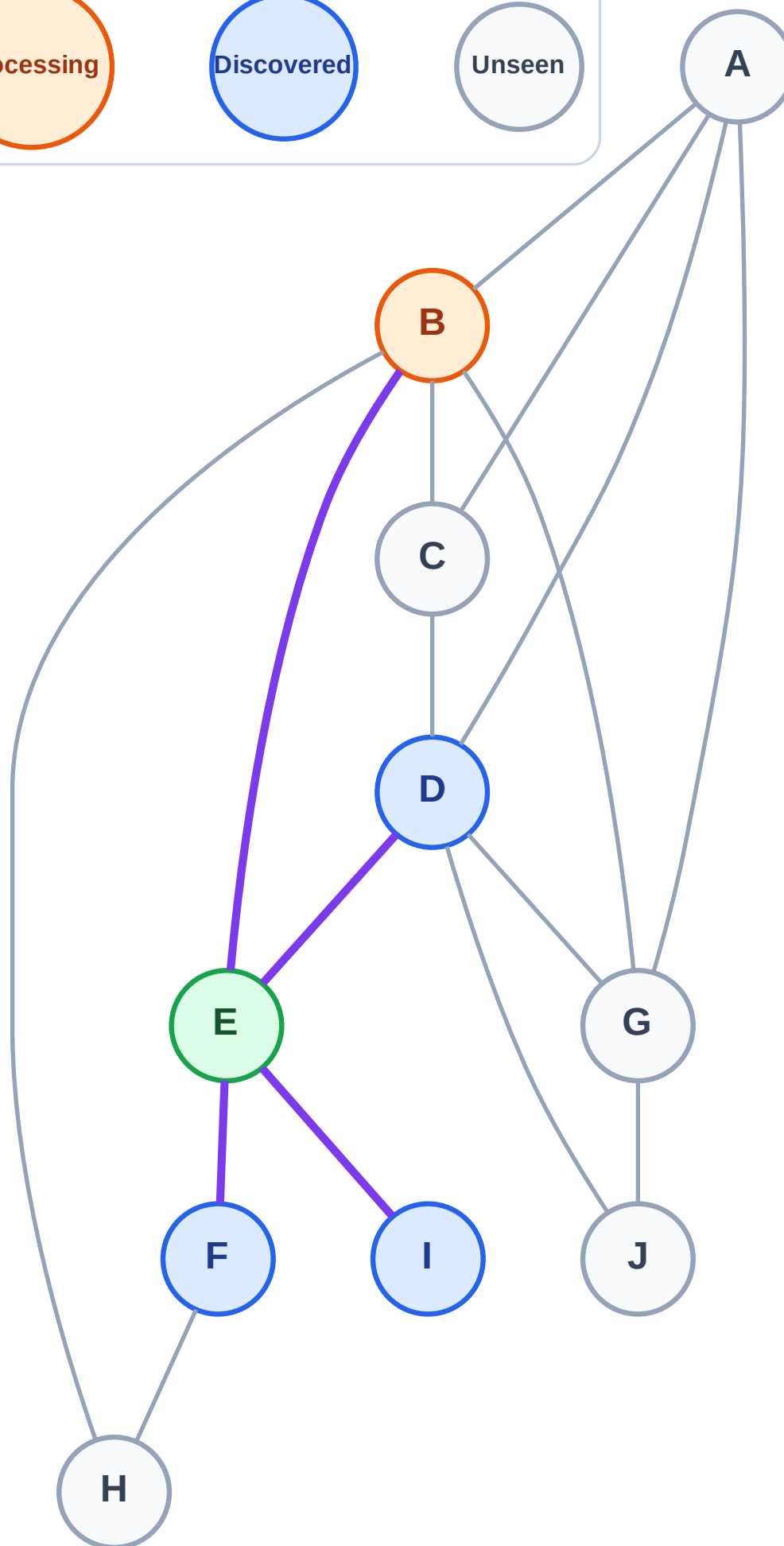
Step 4: Process node B

Legend



Stack (top at right): I | F | D

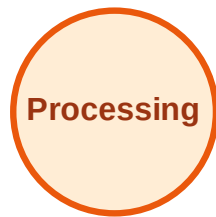
Visited: E



DFS Traversal

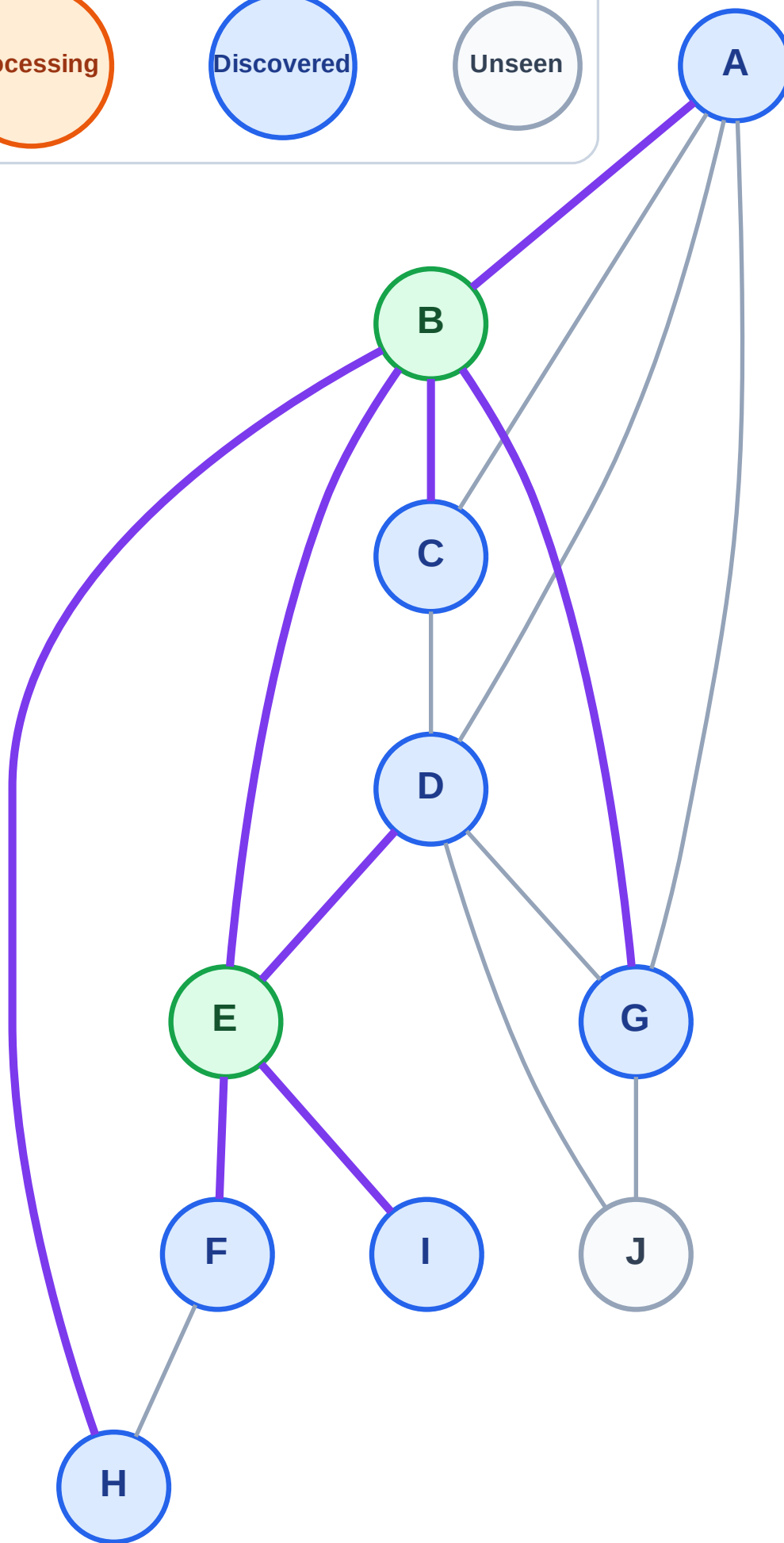
Step 5: Discover H, G, C, A from B

Legend



Stack (top at right): I | F | D | H | G | C | A

Visited: B, E



DFS Traversal

Step 6: Process node A

Legend

Complete

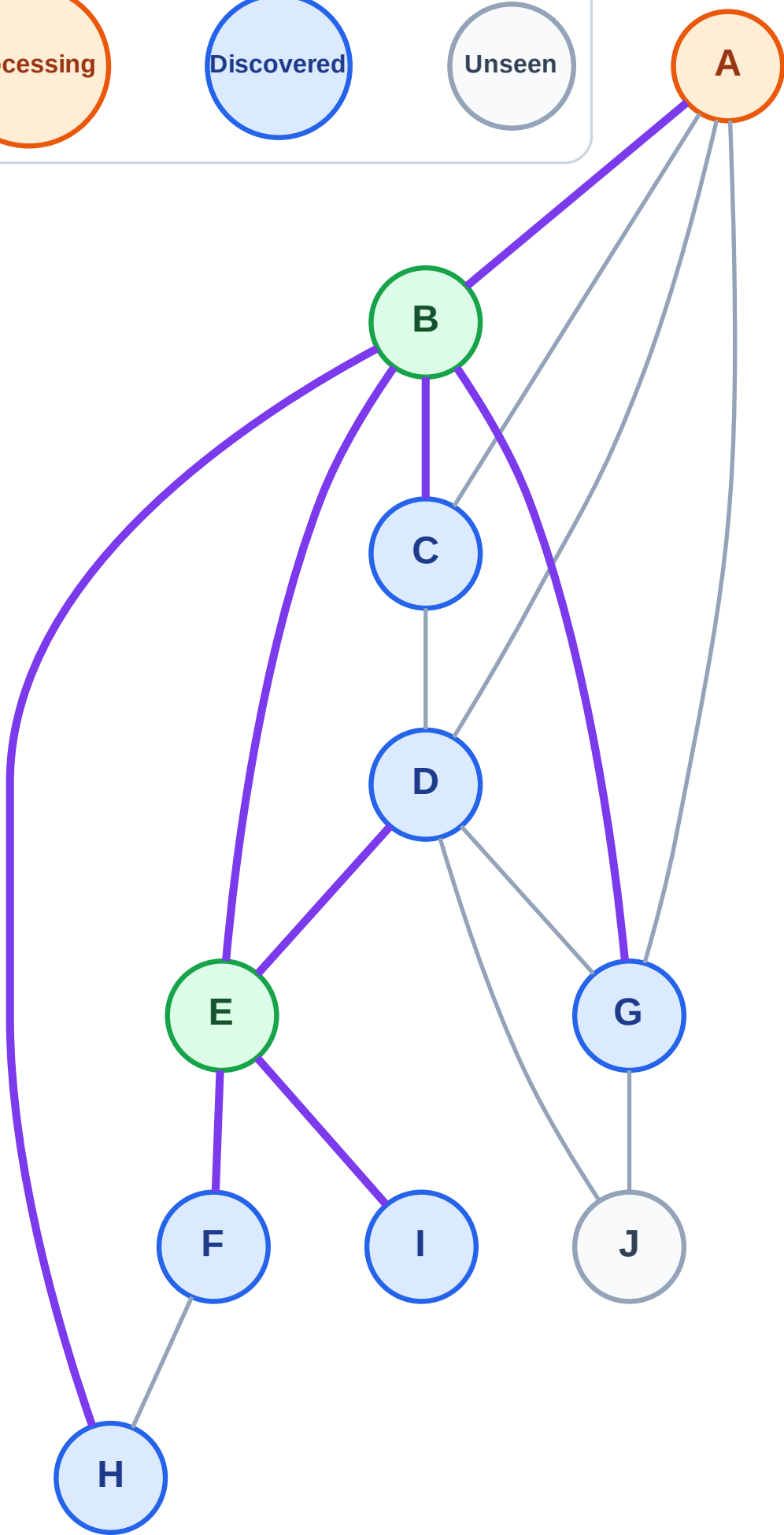
Processing

Discovered

Unseen

Stack (top at right): I | F | D | H | G | C

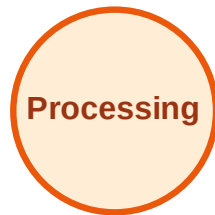
Visited: B, E



DFS Traversal

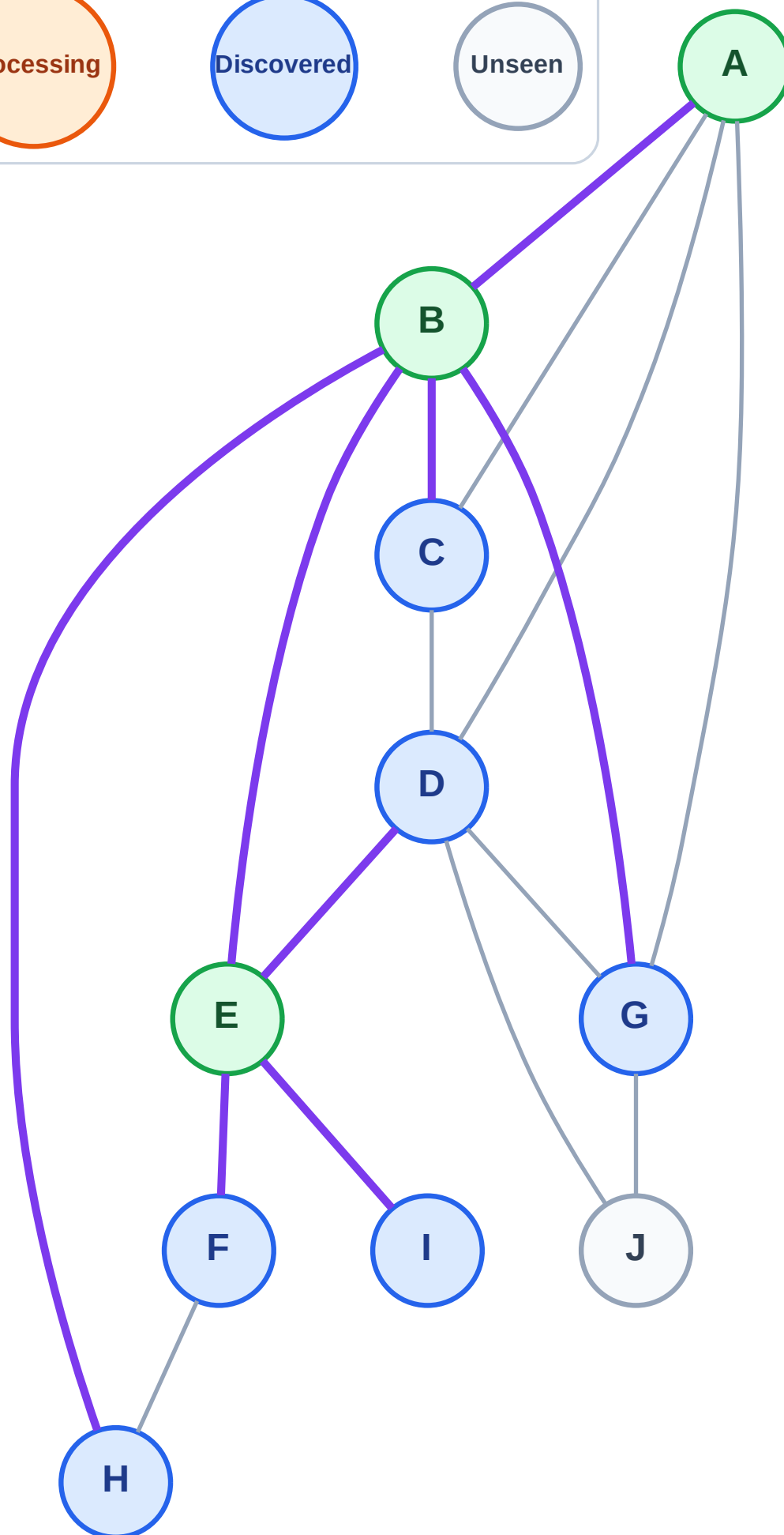
Step 7: No new neighbors from A

Legend



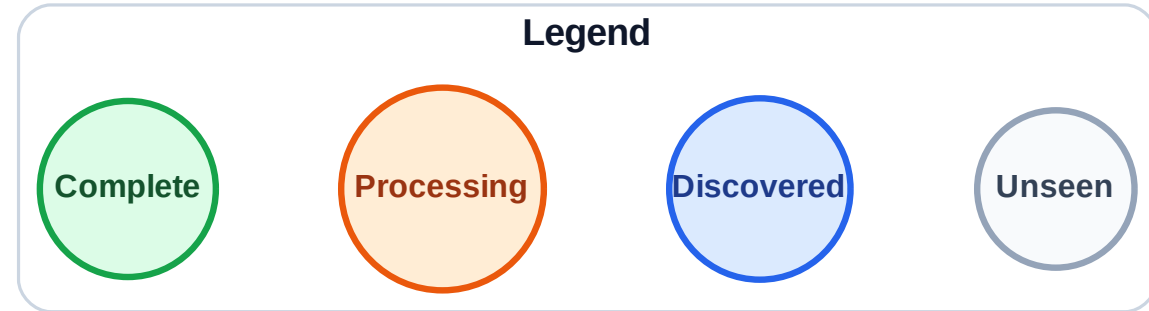
Stack (top at right): I | F | D | H | G | C

Visited: A, B, E



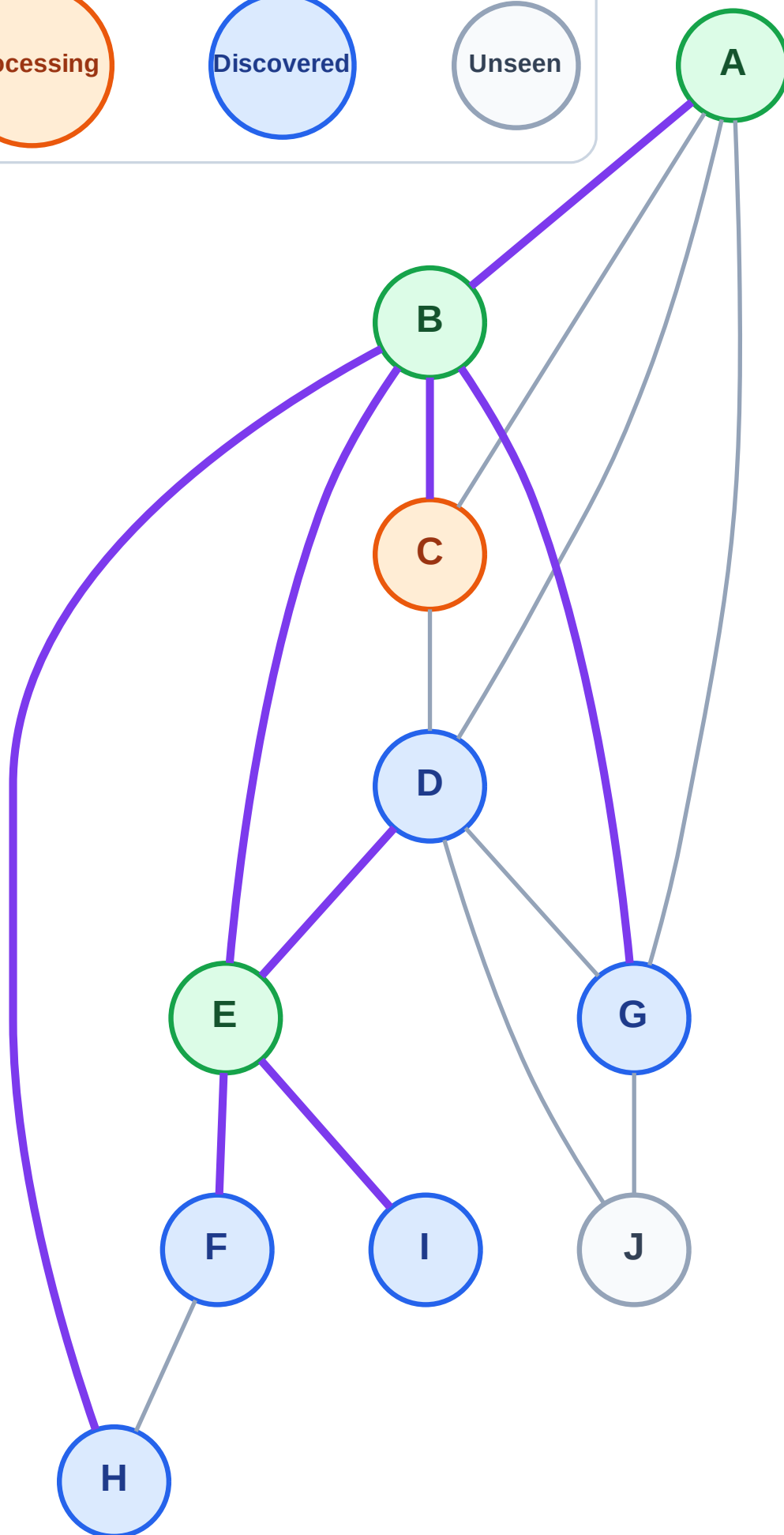
DFS Traversal

Step 8: Process node C



Stack (top at right): I | F | D | H | G

Visited: A, B, E



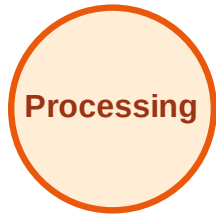
DFS Traversal

Step 9: No new neighbors from C

Legend



Complete



Processing



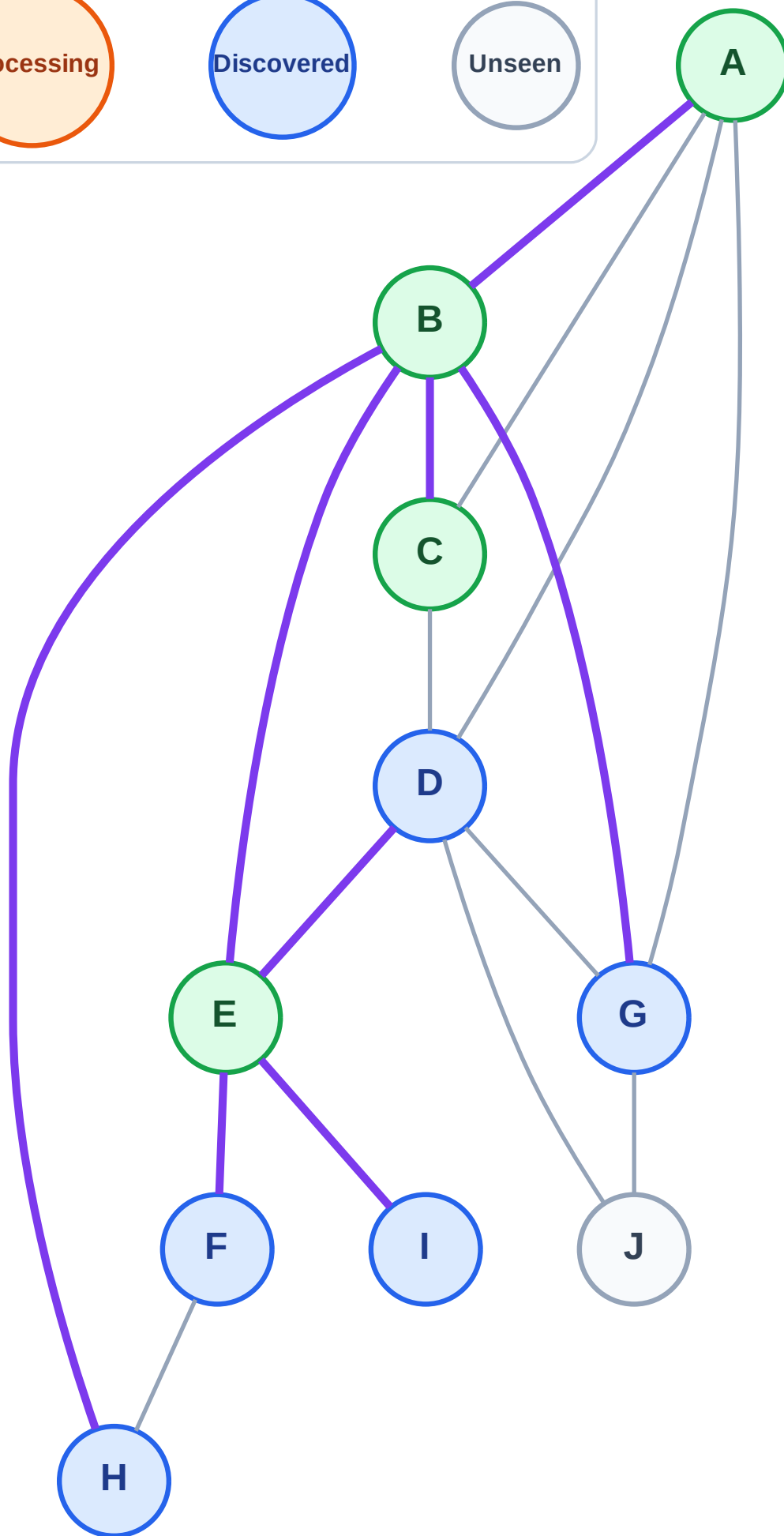
Discovered



Unseen

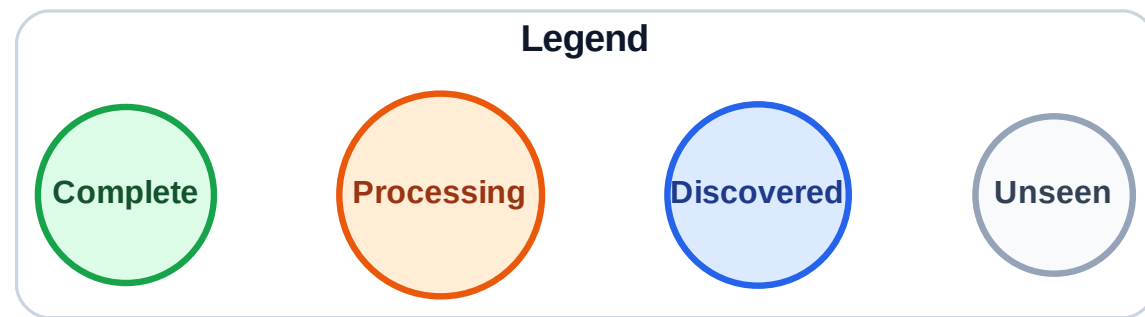
Stack (top at right): I | F | D | H | G

Visited: A, B, C, E

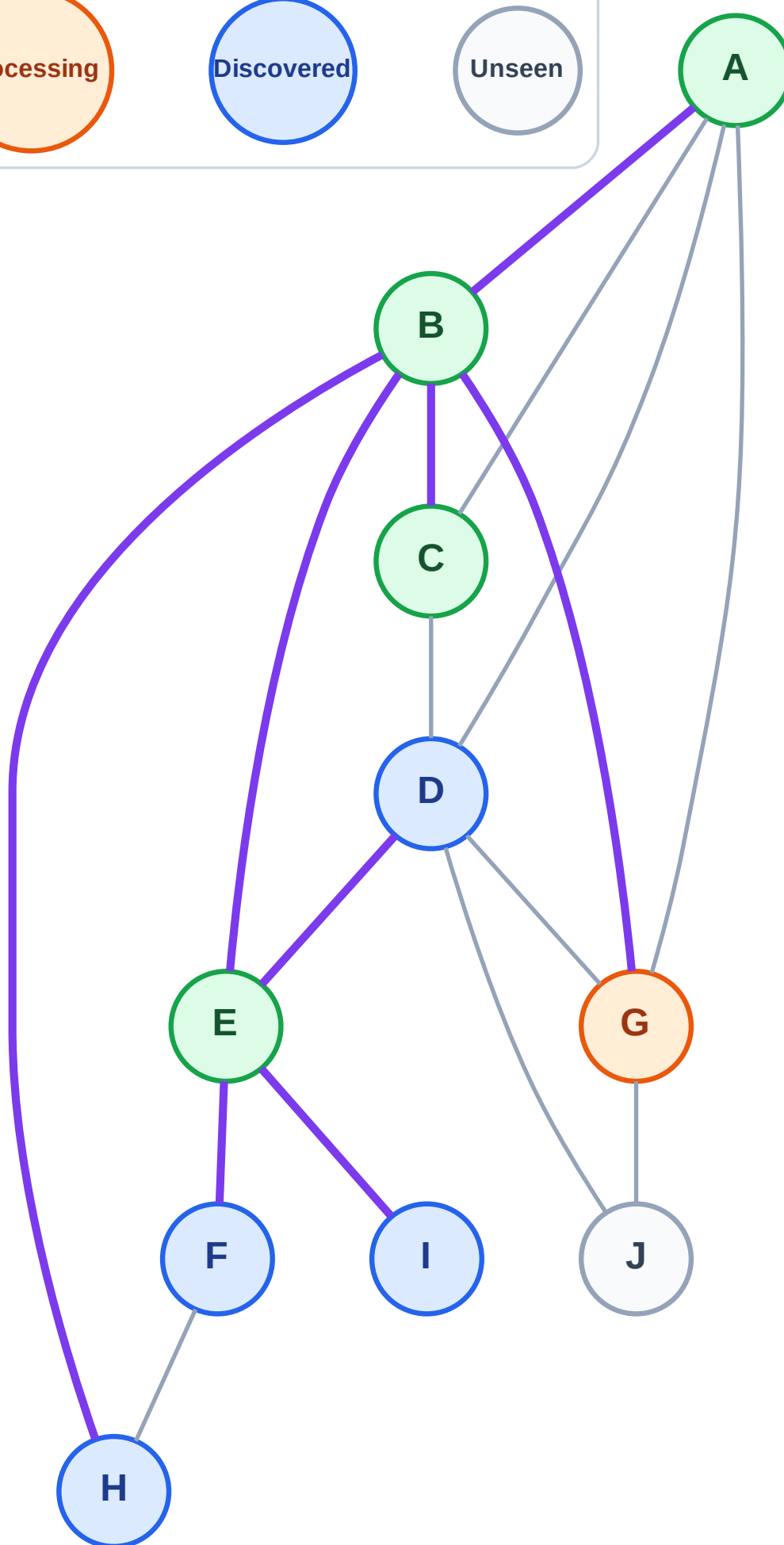


DFS Traversal

Step 10: Process node G



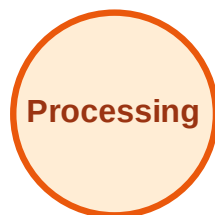
Stack (top at right): I | F | D | H
Visited: A, B, C, E



DFS Traversal

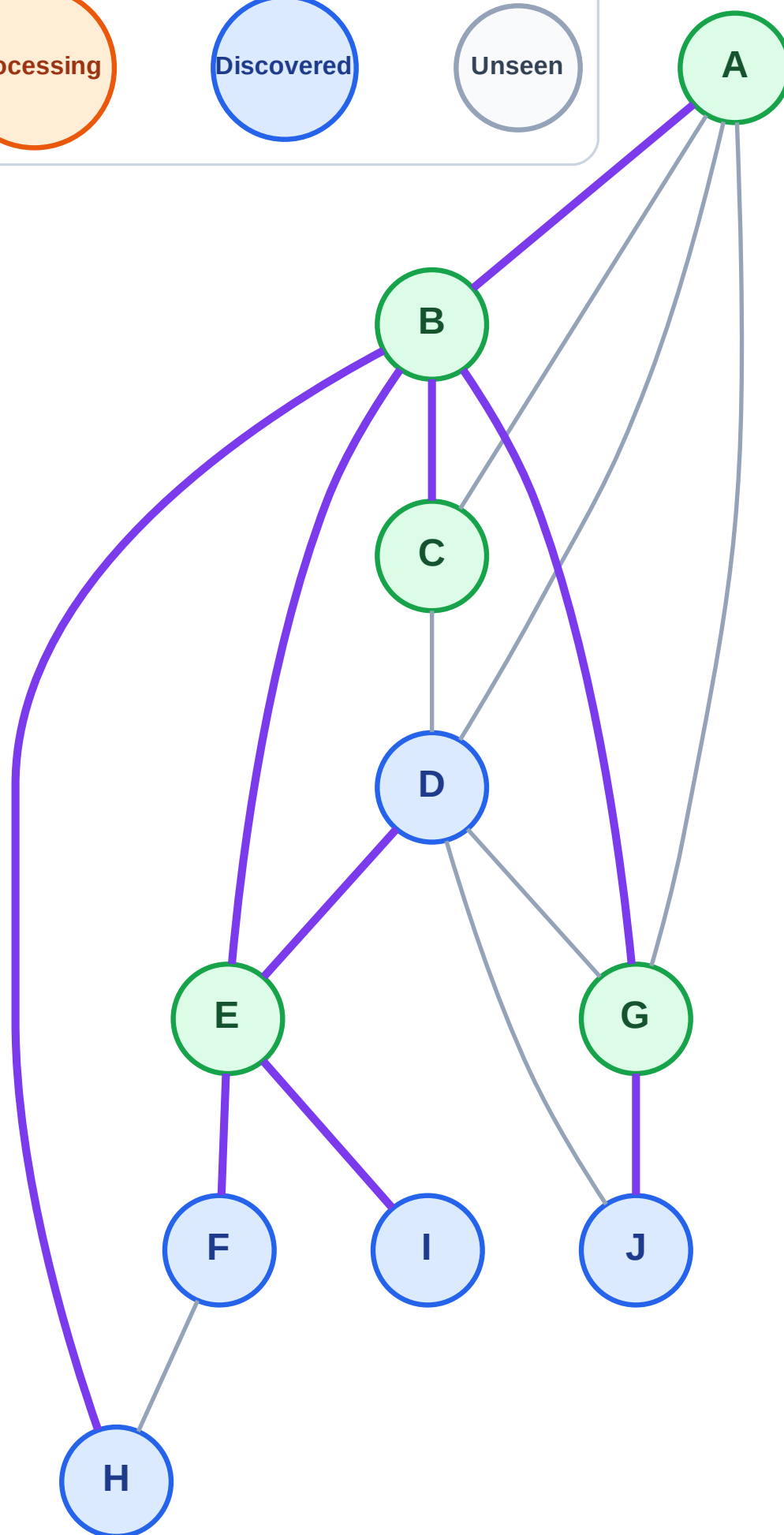
Step 11: Discover J from G

Legend



Stack (top at right): I | F | D | H | J

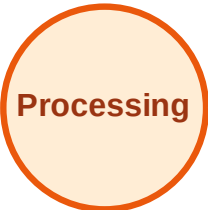
Visited: A, B, C, E, G



DFS Traversal

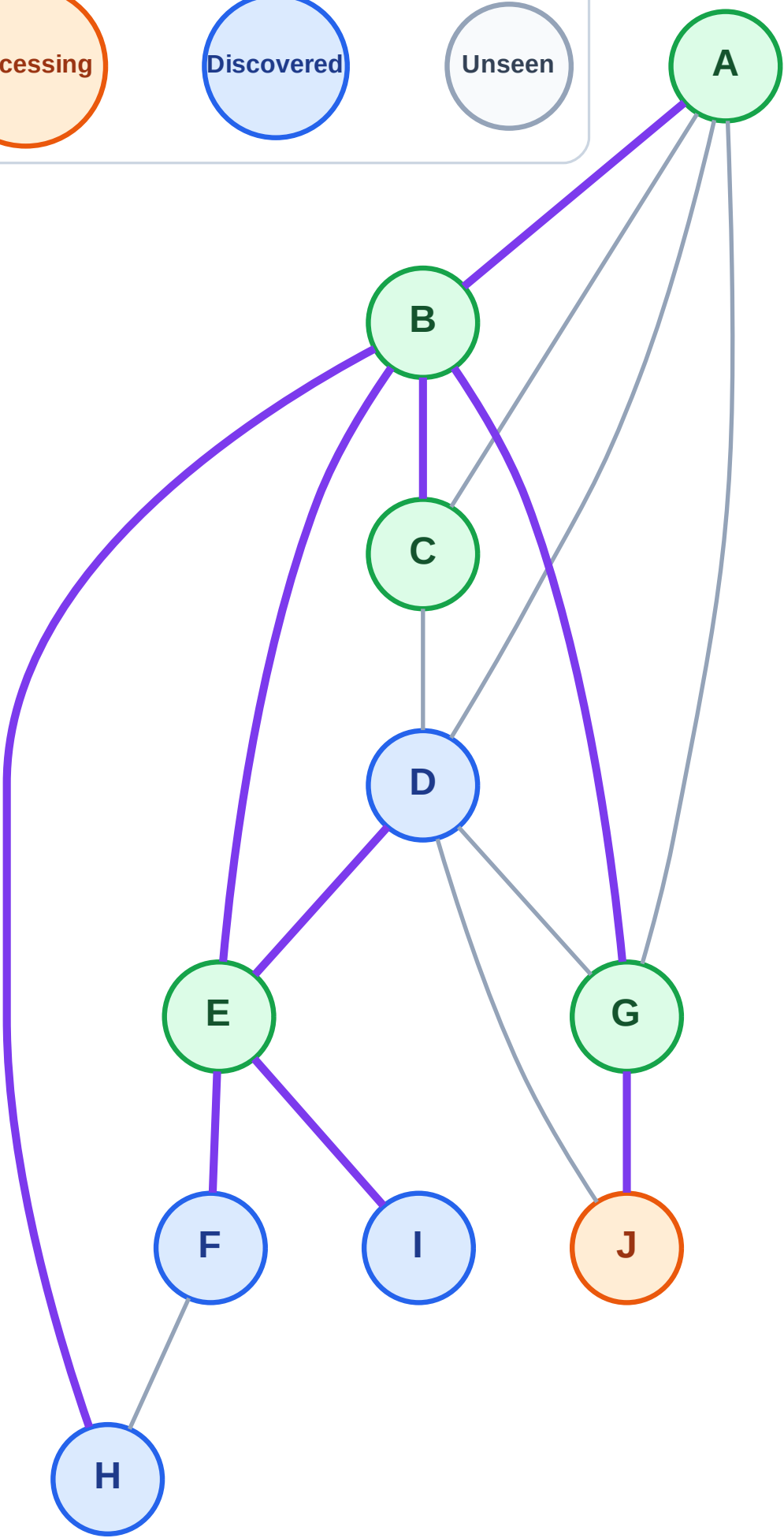
Step 12: Process node J

Legend



Stack (top at right): I | F | D | H

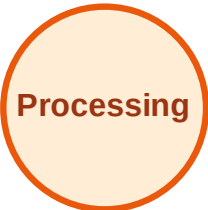
Visited: A, B, C, E, G



DFS Traversal

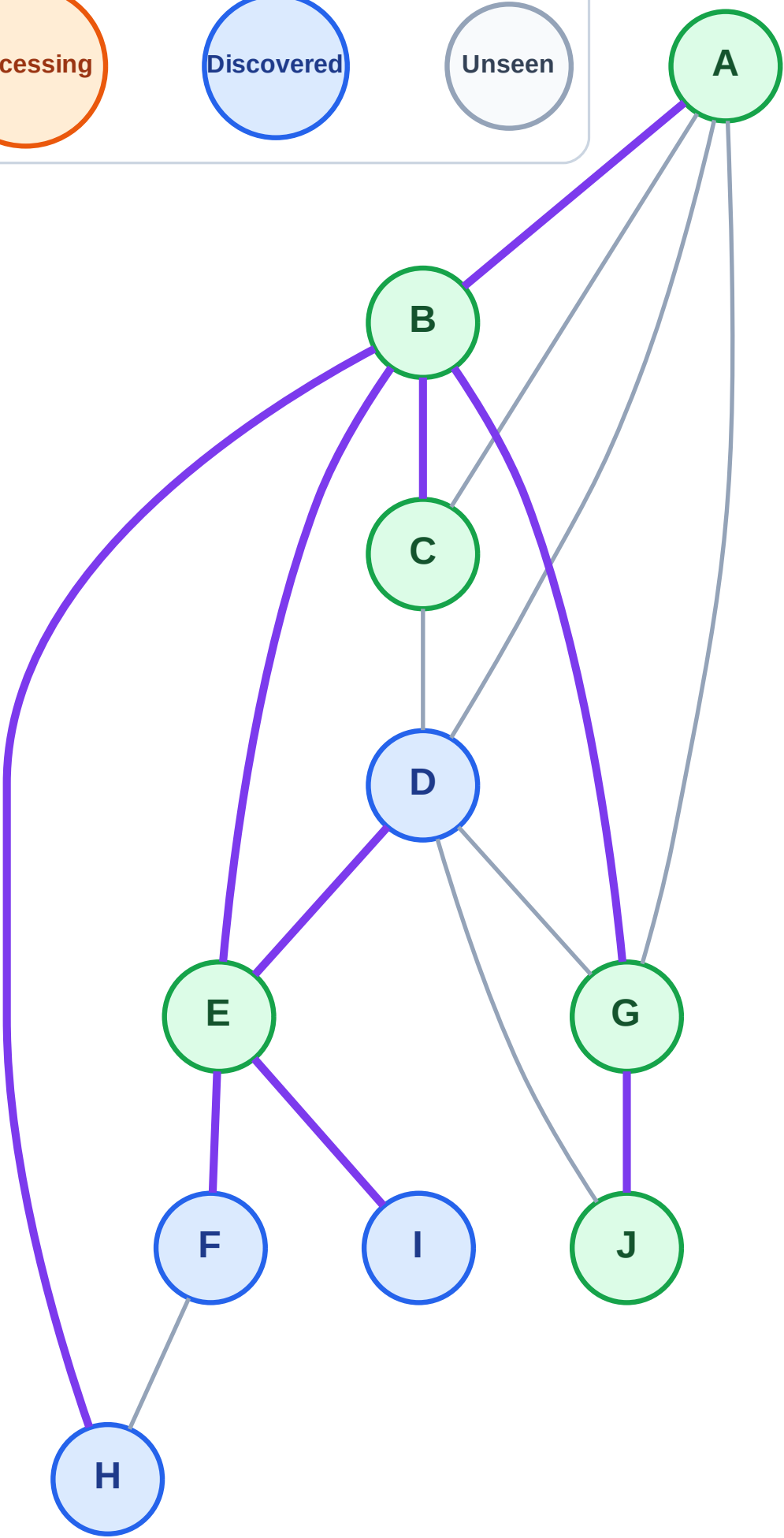
Step 13: No new neighbors from J

Legend



Stack (top at right): I | F | D | H

Visited: A, B, C, E, G, J



DFS Traversal

Step 14: Process node H

Legend

Complete

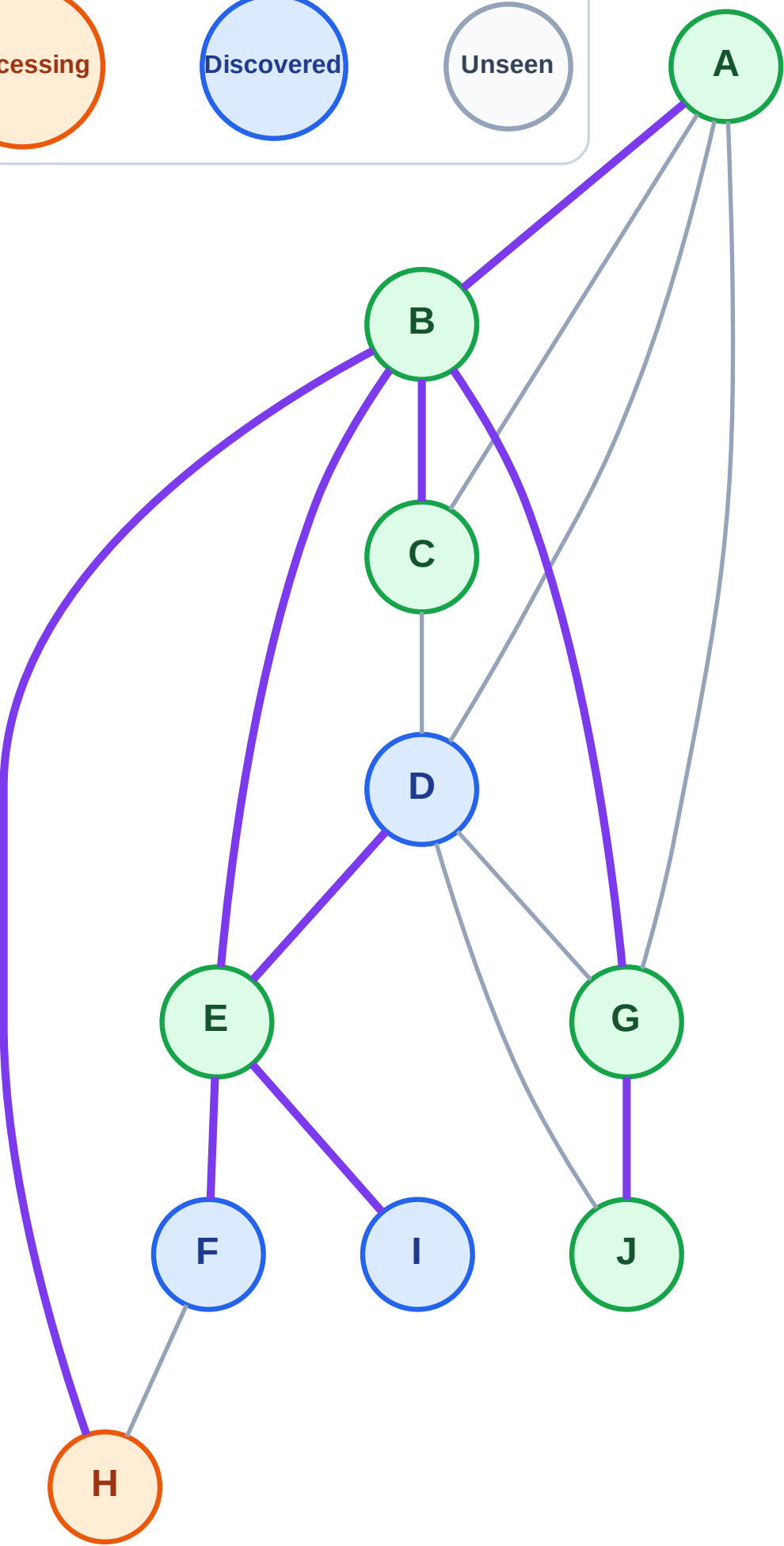
Processing

Discovered

Unseen

Stack (top at right): I | F | D

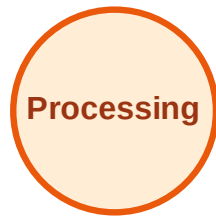
Visited: A, B, C, E, G, J



DFS Traversal

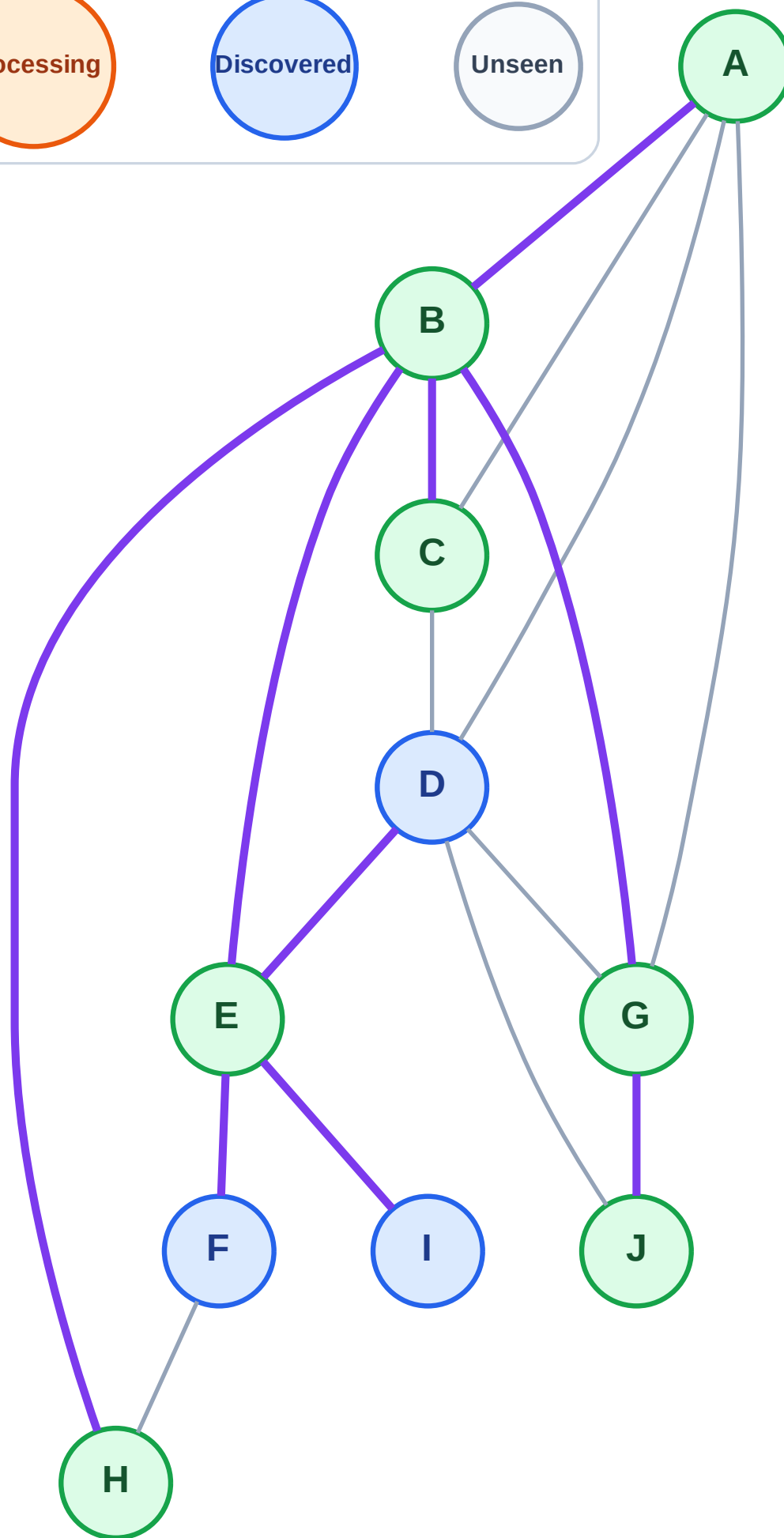
Step 15: No new neighbors from H

Legend



Stack (top at right): I | F | D

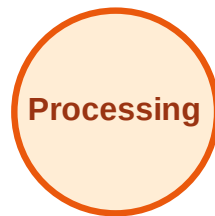
Visited: A, B, C, E, G, H, J



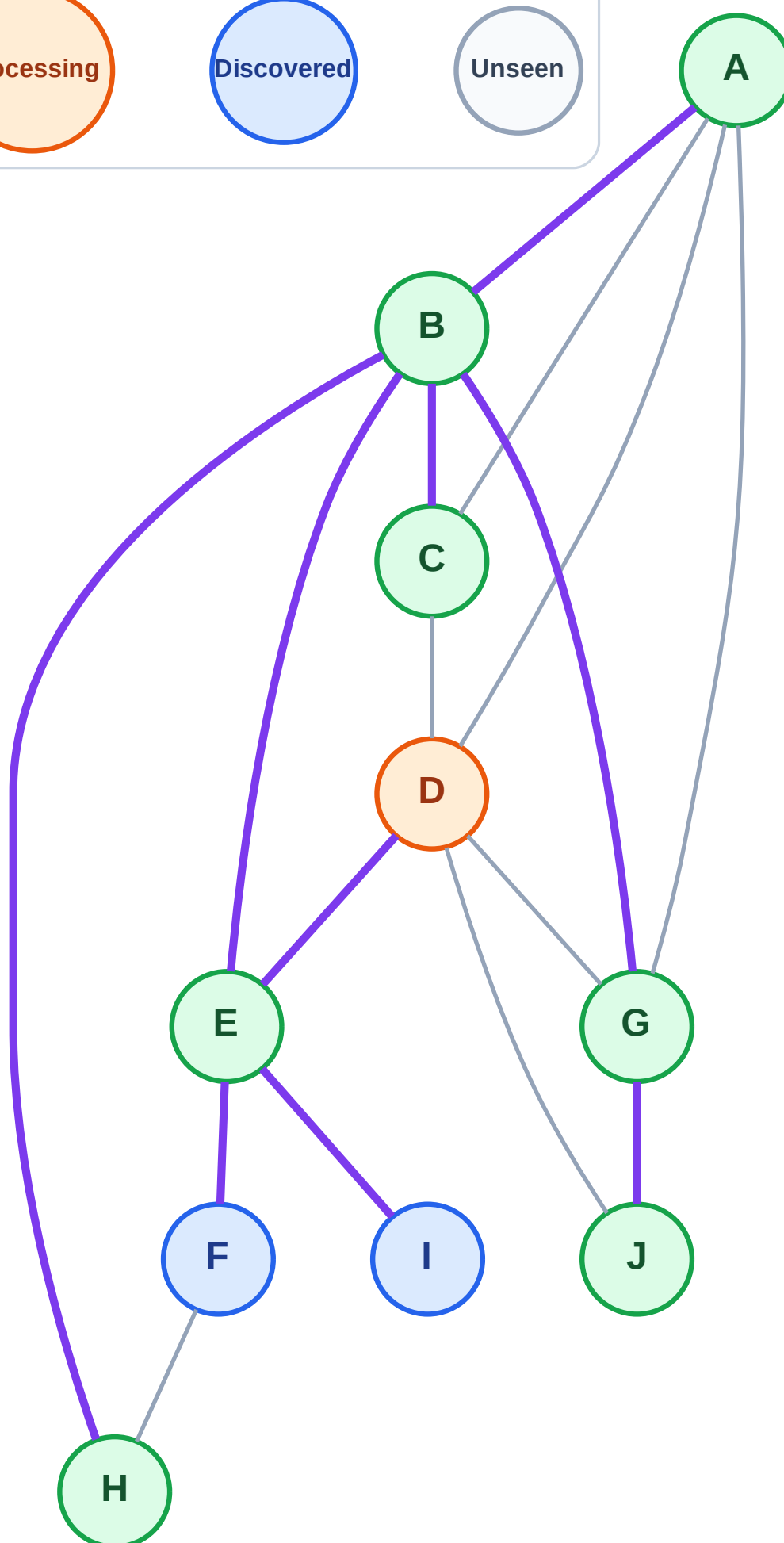
DFS Traversal

Step 16: Process node D

Legend



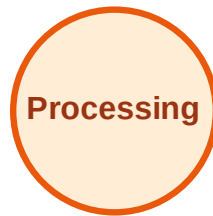
Stack (top at right): I | F
Visited: A, B, C, E, G, H, J



DFS Traversal

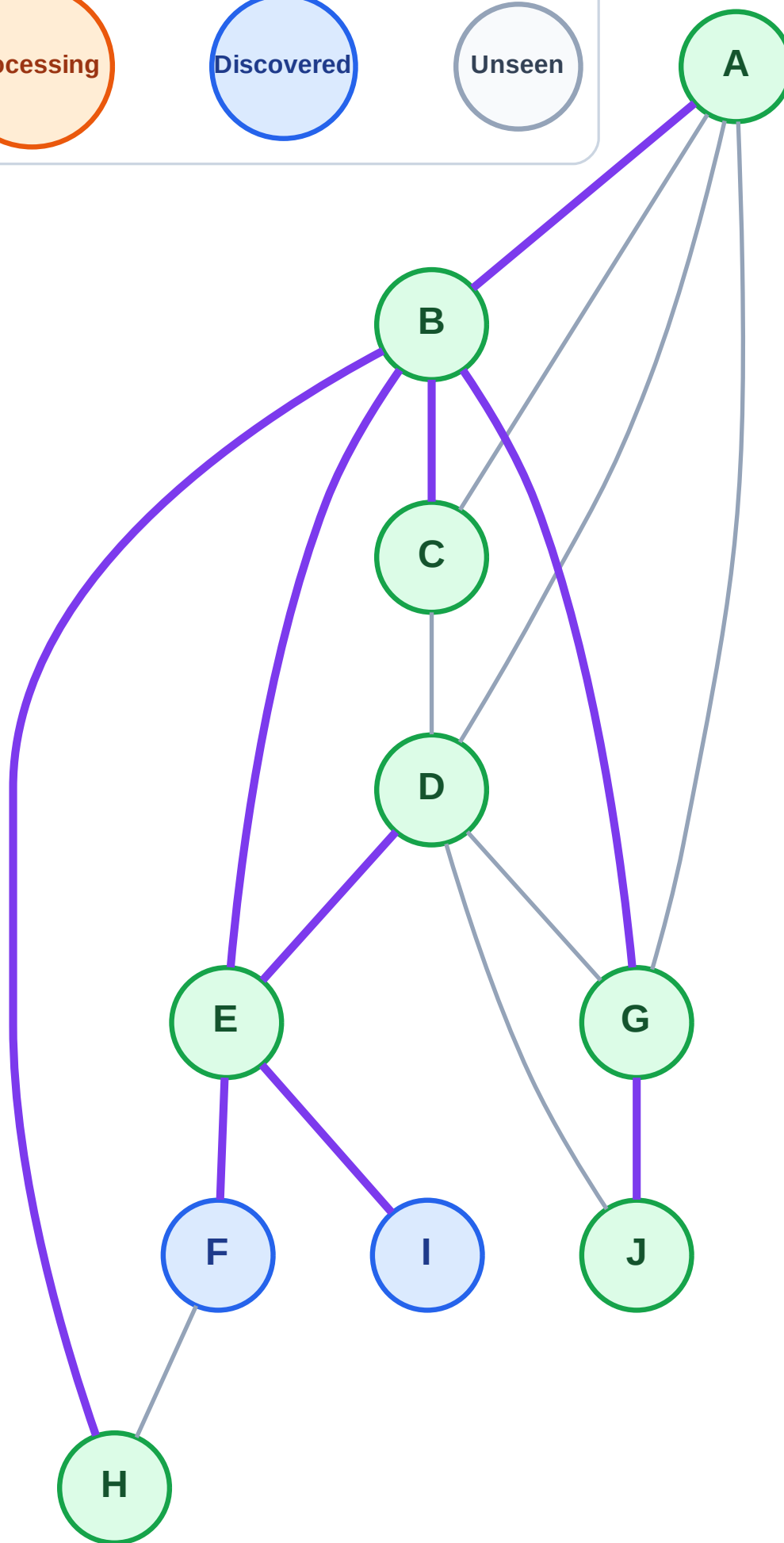
Step 17: No new neighbors from D

Legend



Stack (top at right): I | F

Visited: A, B, C, D, E, G, H, J



DFS Traversal

Step 18: Process node F

Legend

Complete

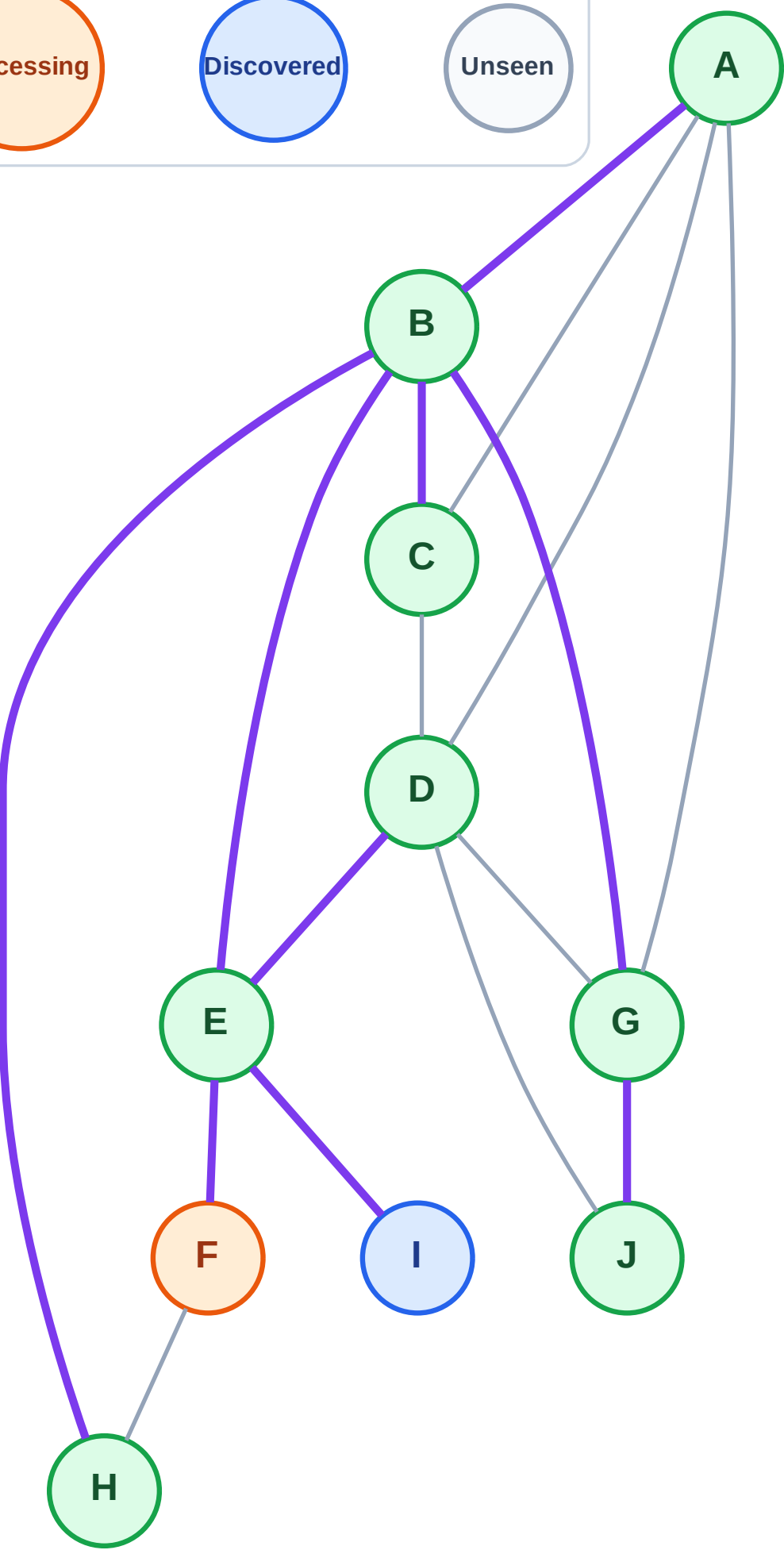
Processing

Discovered

Unseen

Stack (top at right): I

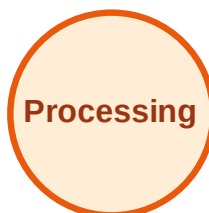
Visited: A, B, C, D, E, G, H, J



DFS Traversal

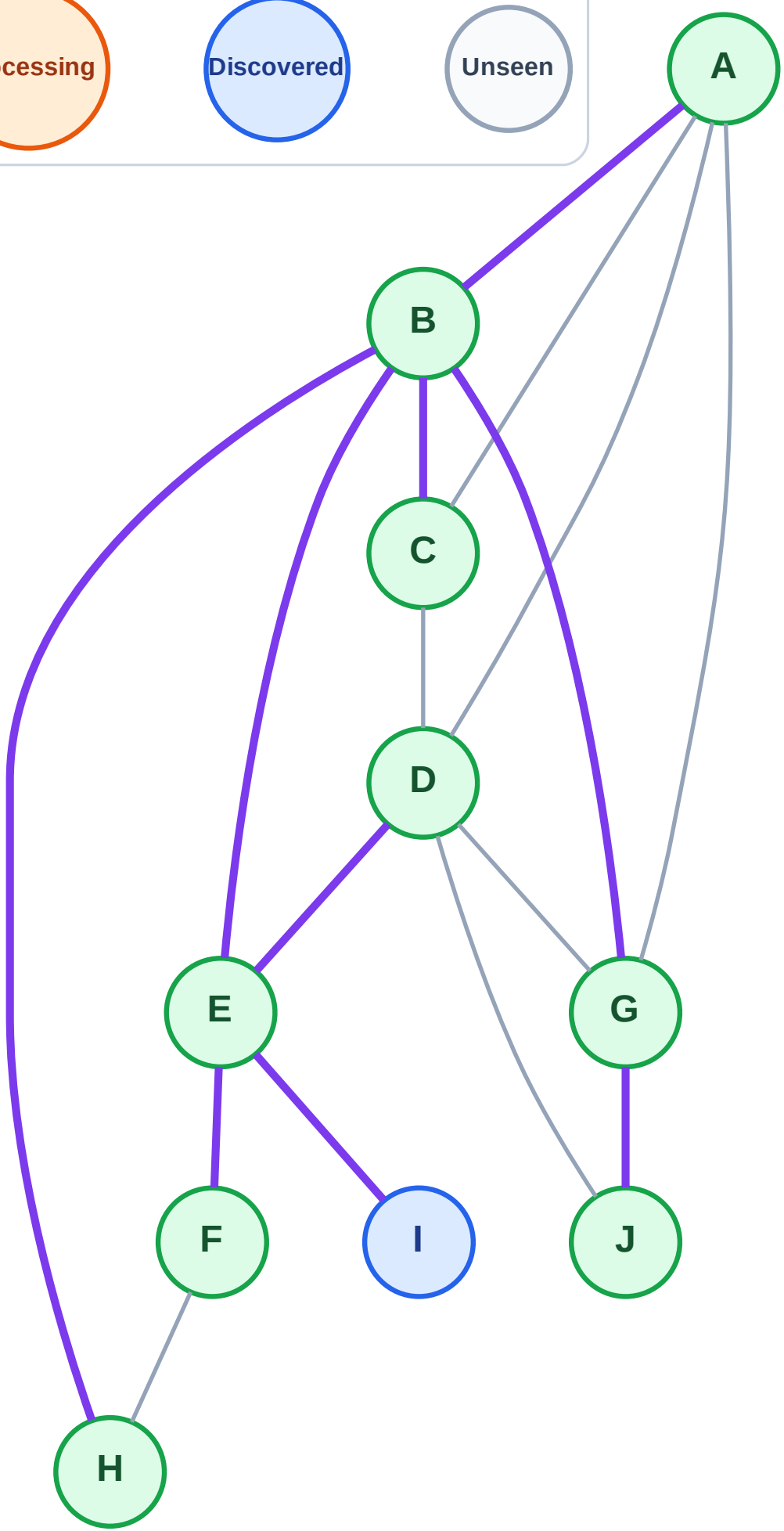
Step 19: No new neighbors from F

Legend



Stack (top at right): I

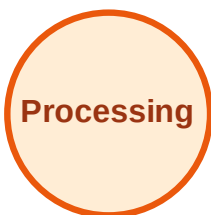
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

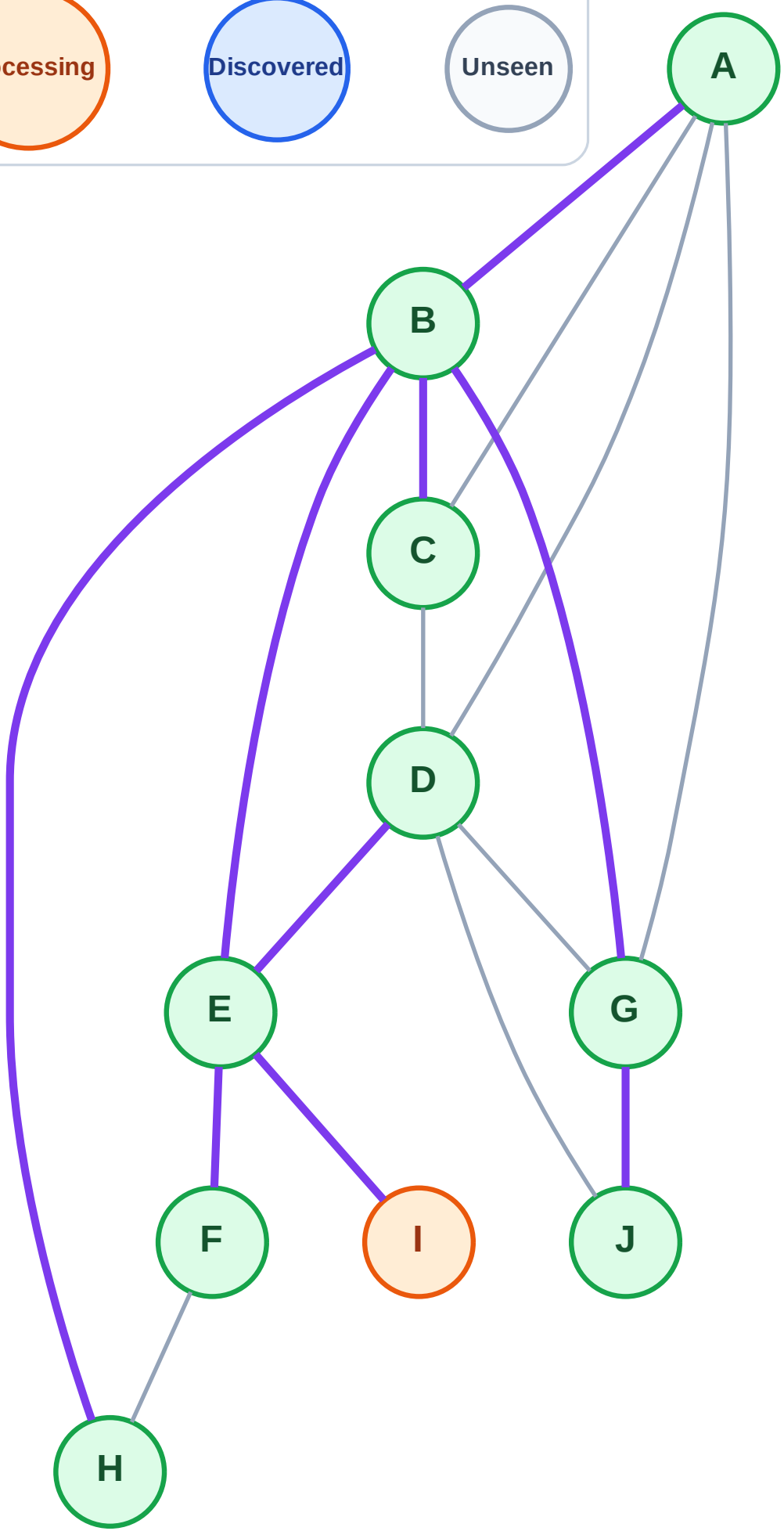
Step 20: Process node I

Legend



Stack (top at right): (empty)

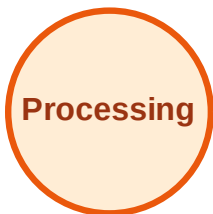
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

